31/7 equals

A. 4 + 1/7

B. 4 + 3/7

C. 3 + 4/7

D. 4 + 3/31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q4/5

Write 100/9 in the form q + r/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.B.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 2

Multiply to check a division • High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1

Compute $(x - 1)(x^2 + 3x - 2) + 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q1/5

Compute $(x^2 + 1)(2x - 3) + (2x - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q2/5

True or false?

$(2x + 1)(3x - 1) - 3 = 6x^2 + x - 4.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q3/5

$(x + 3)(x^2 - 3x + 5)$ equals

A. $x^3 - 4x + 15$

B. $x^3 + 15$

C. $x^3 - 4x + 5$

D. $x^3 + 6x^2 - 4x + 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q4/5

Compute $(x - 2)(x^3 + 2x^2 + 4x + 8)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

High School — Algebra I • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.A.1 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSA-APR.D.6

BACKWARD 3

Use the Remainder Theorem to check the remainder • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2

Find the remainder of $x^3 + 2x^2 - 5x + 3$ divided by $x - 1$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q1/5

Find the remainder of $x^3 - 4x + 5$ divided by $x + 3$ by evaluating.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q2/5

True or false?

The remainder of $x^4 - 16$ divided by $x - 2$ is 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q3/5

The remainder of $6x^2 + x - 4$ divided by $2x + 1$ is

A. 3

B. -4

C. 1

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q4/5

Find the remainder of $2x^3 + x - 1$ divided by $x - 2$.

Remainder: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.2 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSA-APR.D.6

FORWARD 1

Divide with unknown coefficients and higher-degree divisors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

When $x^3 + ax^2 + 2x + 4$ is divided by $x^2 - x + 1$, the remainder is a constant. Find a and the remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q1/5

Rewrite $(x^4 + x^2 + 1)/(x^2 + x + 1)$ in the form $q(x) + r(x)/b(x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q2/5

True or false?

Dividing a degree-5 polynomial by a degree-2 polynomial gives a quotient of degree 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q3/5

The remainder of a division by $x^2 + 4$ can have degree at most

A. 0

B. 1

C. 2

D. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S22-F1-Q4**

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q4/5

Find the polynomial $a(x)$ for which $a(x)/(x - 2) = x^2 + 1 + 3/(x - 2)$.

$a(x) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSA-APR.D.6

FORWARD 2

Use division to find slant asymptotes • High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d

Use division to find the slant asymptote of $f(x) = (x^2 + 3x + 1)/(x + 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q1/5

Find the slant asymptote of $f(x) = (2x^2 - x - 4)/(x - 1)$.

Asymptote: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q2/5

True or false?

A rational function with numerator degree exactly one more than the denominator degree has a slant asymptote.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q3/5

The slant asymptote of $y = (x^2 + 1)/x$ is

A. $y = x$

B. $y = 1$

C. $x = 0$

D. $y = x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q4/5

Explain why the graph of $y = x + 2 - 1/(x + 1)$ gets close to $y = x + 2$ for large $|x|$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Interpreting Functions • HSF-IF.C.7.d • Set 22 • FORWARD 2 • Q5/5

Rewrite simple rational expressions by inspection

MAIN	Algebra II main question	**HSA-APR.D.6**
BACKWARD 1	Split a fraction with a sum in the numerator	**7.EE.A.1**
BACKWARD 2	Rewrite a numerator as a multiple of the denominator plus a cons…	**6.EE.A.3**
BACKWARD 3	Divide by long division when inspection fails	**HSA-APR.D.6**
FORWARD 1	Use the rewritten form to reason about values	**HSA-APR.D.6**
FORWARD 2	Graph rational functions as transformations of y = 1/x	**HSF-BF.B.3**

SET 23 • HSA-APR.D.6

MAIN

Algebra II main question • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Rewrite $(x + 5)/(x + 2)$ by inspection in the form $q + r/(x + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q1/5

Rewrite $(2x - 1)/(x - 3)$ by inspection in the form $q + r/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q2/5

Rewrite $(x^2 + 1)/x$ by inspection in the form $q(x) + r/x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q3/5

Rewrite $(3x + 7)/(x + 1)$ by inspection in the form $q + r/(x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q4/5

Rewrite $x^2/(x - 1)$ by inspection in the form $q(x) + r/(x - 1)$. (Hint: $x^2 = (x^2 - 1) + 1$.)

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • MAIN • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 1

Split a fraction with a sum in the numerator • Grade 7 • Expressions & Equations • 7.EE.A.1

Write (x + 3)/x as a sum of two fractions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q1/5

Simplify (2(x − 3) + 5)/(x − 3) as a sum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q2/5

True or false?

$(a + b)/c = a/c + b/c$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q3/5

$(x^2 + 1)/x$ equals

A. $x + 1/x$

B. $1 + 1/x$

C. $x + 1$

D. $x^2 + 1/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q4/5

True or false?

$c/(a + b) = c/a + c/b$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 2

Rewrite a numerator as a multiple of the denominator plus a constant • Grade 6 • Expressions & Equations • 6.EE.A.3

Find c so that $x + 5 = (x + 2) + c$.

$c =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q1/5

Find c so that $2x - 1 = 2(x - 3) + c$.

c = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q2/5

True or false?

$3x + 7 = 3(x + 1) + 7.$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q3/5

$x^2 - 1$ equals

A. $x(x - 1)$

B. $(x + 1)^2$

C. $(x - 1)^2$

D. $(x - 1)(x + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q4/5

Find c so that $4x + 1 = 4(x - 2) + c$.

$c =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • HSA-APR.D.6

BACKWARD 3

Divide by long division when inspection fails • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

Divide x^2 by $x - 1$ with long division. Give the quotient and remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q1/5

Divide $3x + 7$ by $x + 1$. Give the quotient and remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q2/5

True or false?

Long division and inspection give the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q3/5

The remainder of $(2x - 1) \div (x - 3)$ is

A. -5

B. 2

C. -1

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q4/5

Divide x + 5 by x + 2. Give the quotient and remainder.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • HSA-APR.D.6

FORWARD 1

Use the rewritten form to reason about values • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6

For which integers x is $(x + 5)/(x + 2)$ an integer? Use $1 + 3/(x + 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q1/5

Explain why $(2x - 1)/(x - 3)$ can never equal 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q2/5

True or false?

$(3x + 7)/(x + 1) > 3$ for every $x > -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q3/5

As x grows large, (x + 5)/(x + 2) approaches

A. 3

B. 0

C. 5/2

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q4/5

For $x > 0$, what is the least value of $x + 1/x$, and where does it occur?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.6 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSA-APR.D.6

FORWARD 2

Graph rational functions as transformations of y = 1/x • High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3

Describe how the graph of $y = 1 + 3/(x + 2)$ is obtained from $y = 3/x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q1/5

Give the asymptotes of $y = 2 + 5/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q2/5

True or false?

$y = (3x + 7)/(x + 1)$ has horizontal asymptote $y = 7$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q3/5

$y = (x + 5)/(x + 2)$ is a translation of $y = 3/x$ by

A. left 2, up 1

B. right 2, up 1

C. left 2, up 5

D. left 5, up 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S23-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q4/5

Write $(4x - 3)/(x - 1)$ as a transformation of k/x and give its asymptotes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Building Functions • HSF-BF.B.3 • Set 23 • FORWARD 2 • Q5/5

Multiply and divide rational expressions

MAIN	Algebra II main question	**HSA-APR.D.7**
BACKWARD 1	Multiply and divide fractions	**6.NS.A.1**
BACKWARD 2	Factor numerators and denominators	**HSA-SSE.A.2**
BACKWARD 3	Division by zero is undefined	**7.NS.A.2.b**
FORWARD 1	Simplify complex fractions	**HSA-APR.D.7**
FORWARD 2	Explain closure of rational expressions	**HSA-APR.D.7**

SET 24 • HSA-APR.D.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify $(x^2 - 9)/(x^2 + x - 6) \cdot (x - 2)/(x - 3)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q1/5

Simplify $(x + 2)/(3x^2) \div (x^2 - 4)/x$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q2/5

Simplify $(2x - 6)/(x + 1) \cdot (x + 1)/(x^2 - 9)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q3/5

Simplify $(3x + 3)/(x - 4) \div (x + 1)/(x - 4)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q4/5

Simplify $(5x^2 + 5x)/(x^2 - 4) \div (x + 1)/(x - 2)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • MAIN • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 1

Multiply and divide fractions • Grade 6 • The Number System • 6.NS.A.1

Compute 2/3 · 9/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q1/5

Compute 5/6 ÷ 10/9.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q2/5

True or false?

$a/b \div c/d = a/b \cdot c/d$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q3/5

3/8 ÷ 3/4 equals

A. 9/32

B. 2

C. 3/4

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q4/5

Compute 14/15 · 5/7.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 2

Factor numerators and denominators • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 + x - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q1/5

Factor $5x^2 + 5x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q2/5

$x^2 - 4$ equals

A. $(x - 4)(x + 1)$

B. $(x - 2)^2$

C. $(x - 2)(x + 2)$

D. $x(x - 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q3/5

True or false?

$2x - 6 = 2(x - 3)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q4/5

Factor $x^2 - 9$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • HSA-APR.D.7

BACKWARD 3

Division by zero is undefined • Grade 7 • The Number System • 7.NS.A.2.b

For which x is 1/(x − 4) undefined?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q1/5

For which x is $(x + 2)/(x^2 - 9)$ undefined?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q2/5

True or false?

After canceling $(x - 3)$, $x = 3$ is allowed in the original expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q3/5

Dividing by $(x + 1)/(x - 2)$ is impossible when

A. $x = -1$ only

B. $x = 0$

C. $x = -1$ or $x = 2$

D. $x = 2$ only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q4/5

Why is $x = 0$ excluded from $(x + 2)/(3x^2)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 24 • BACKWARD 3 • Q5/5

SET 24 • HSA-APR.D.7

FORWARD 1

Simplify complex fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify (1/x + 1)/(1/x − 1). State the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q1/5

Simplify $(1 - 4/x^2)/(1 + 2/x)$. State the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q2/5

True or false?

$(a/b)/(c/d) = ad/(bc)$ when $b, c, d \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q3/5

$(x/2)/(x/4)$ for $x \neq 0$ equals

A. 2

B. 1/2

C. $x^2/8$

D. $8/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q4/5

Simplify $(x - 1/x)/(x + 1)$. State the excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 1 • Q5/5

SET 24 • HSA-APR.D.7

FORWARD 2

Explain closure of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Explain why the product of two rational expressions is a rational expression.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q1/5

Explain why division by a nonzero rational expression gives a rational expression.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q2/5

True or false?

Rational expressions are closed under division by the zero expression.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q3/5

Which system is closed under addition, subtraction, multiplication, and division by nonzero elements?

A. Whole numbers

B. Rational expressions

C. Integers

D. Polynomials

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q4/5

Give an example showing polynomials are not closed under division.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 24 • FORWARD 2 • Q5/5

Add and subtract rational expressions

MAIN	Algebra II main question	**HSA-APR.D.7**
BACKWARD 1	Add fractions with unlike denominators	**5.NF.A.1**
BACKWARD 2	Find a least common denominator by factoring	**HSA-SSE.A.2**
BACKWARD 3	Subtract a whole numerator	**6.EE.A.3**
FORWARD 1	Decompose into partial fractions	**HSA-APR.D.7**
FORWARD 2	Solve rational equations	**HSA-REI.A.2**

SET 25 • HSA-APR.D.7

MAIN

Algebra II main question • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Simplify $(x + 1)/(x^2 + x) + 1/x$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q1/5

Write $\frac{2}{(x + 1)} + \frac{3}{(x - 2)}$ as one rational expression. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q2/5

Simplify $x/(x-3) - 3/(x-3)$. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q3/5

Write $1/(x^2 - 1) - 1/(x - 1)$ as one rational expression. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q4/5

Write $4/(x^2 - 4x) - 1/x$ as one rational expression. State all excluded values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • MAIN • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 1

Add fractions with unlike denominators • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Compute 2/3 + 3/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q1/5

Compute 5/6 − 1/4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q2/5

True or false?

1/2 + 1/3 = 2/5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q3/5

1/4 + 1/6 equals

A. 5/12

B. 2/10

C. 2/12

D. 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q4/5

Compute 3/10 + 2/15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.A.1 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 2

Find a least common denominator by factoring • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Factor $x^2 + x$ and $x^2 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q1/5

What is the least common denominator of $1/(x^2 - 1)$ and $1/(x - 1)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q2/5

The LCD of $1/x$ and $1/(x(x-4))$ is

A. $x - 4$

B. $x^2(x - 4)$

C. $x(x - 4)$

D. x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q3/5

True or false?

The LCD of $2/(x + 1)$ and $3/(x - 2)$ is $x - 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q4/5

What is the LCD of $1/(x^2 + x)$ and $1/x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • HSA-APR.D.7

BACKWARD 3

Subtract a whole numerator • Grade 6 • Expressions & Equations • 6.EE.A.3

Simplify $4 - (x - 4)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q1/5

Simplify $1 - (x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q2/5

True or false?

$3 - (x - 3) = -x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q3/5

$2(x - 2) - 3(x + 1)$ equals

A. $5x - 7$

B. $-x + 1$

C. $-x - 7$

D. $-x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q4/5

Simplify $x - (3 - x)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 25 • BACKWARD 3 • Q5/5

SET 25 • HSA-APR.D.7

FORWARD 1

Decompose into partial fractions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

Find A and B so that

$$\frac{5x + 1}{(x - 1)(x + 2)} = \frac{A}{x - 1} + \frac{B}{x + 2}.$$

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q1/5

Find A and B so that $1/(x(x + 1)) = A/x + B/(x + 1)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q2/5

True or false?

$1/(x(x + 1)) = 1/x + 1/(x + 1)$ for $x \neq 0, -1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q3/5

$4/(x^2 - 4)$ equals

A. $2/(x - 2) - 2/(x + 2)$

B. $1/(x - 2) + 1/(x + 2)$

C. $4/(x - 2) - 4/(x + 2)$

D. $1/(x - 2) - 1/(x + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S25-F1-Q4**

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q4/5

Use $1/(n(n+1)) = 1/n - 1/(n+1)$ to find $1/(1\cdot2) + 1/(2\cdot3) + 1/(3\cdot4) + 1/(4\cdot5)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 25 • FORWARD 1 • Q5/5

SET 25 • HSA-APR.D.7

FORWARD 2

Solve rational equations • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $2/(x + 1) + 3/(x - 2) = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q1/5

Solve $x/(x-3) - 3/(x-3) = 1$. Explain your answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q2/5

True or false?

A solution of a rational equation that makes a denominator 0 must be rejected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q3/5

The solution of $1/x + 1/(2x) = 3$ is

A. $x = 1/2$

B. $x = 2$

C. $x = 3/2$

D. no solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q4/5

Solve $4/(x^2 - 4x) - 1/x = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 25 • FORWARD 2 • Q5/5

SET 26 • HSA-CED.A.1 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Create and solve rational, radical, and exponential equations from contexts

MAIN	Algebra II main question	**HSA-CED.A.1**
BACKWARD 1	Express rates of work and unit rates	**6.RP.A.3.b**
BACKWARD 2	Write exponential growth models	**HSF-LE.A.2**
BACKWARD 3	Solve linear equations after clearing a denominator	**HSA-REI.B.3**
FORWARD 1	Create equations whose algebraic solutions must be checked aga.	**HSA-CED.A.1**
FORWARD 2	Solve exponential context equations with logarithms	**HSF-LE.A.4**

SET 26 • HSA-CED.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

Ana can paint a room in 4 hours. Working together, Ana and Ben paint it in 2.4 hours. Write an equation for Ben's time b (in hours) working alone, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q1/5

A car's speed s (in mph) before braking is estimated from its skid length d (in feet) by $s = \sqrt{(22.5d)}$. Write and solve an equation for the skid length of a car going 45 mph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q2/5

A town of 1,200 people doubles in population every 5 years. Write an equation for the time t (in years) when the population is 9,600, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q3/5

A printer costs $500 plus $4 per shirt. Write an equation for the number of shirts x that makes the average cost per shirt $9, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q4/5

A tank holds 10 L of a 20% salt solution. Write an equation for the liters x of pure water to add to make an 8% solution, and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • MAIN • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 1

Express rates of work and unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana paints a room in 4 hours. What fraction of the room does she paint per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q1/5

A pump fills a pool in 6 hours. What fraction does it fill in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q2/5

True or false?

If two people work together, their times to finish the job add.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q3/5

A machine makes 300 parts in 5 hours. Its rate is

A. 1500 parts per hour

B. 300 parts per hour

C. 60 parts per hour

D. 5/300 parts per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q4/5

A job takes 2.4 hours together. What fraction is done per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 26 • BACKWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 2

Write exponential growth models • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

A population of 1,200 doubles every 5 years. Write the population after t years.

P(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q1/5

A $500 deposit grows 3% per year. Write its value after t years.

V(t) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q2/5

True or false?

A quantity that doubles every 5 years is multiplied by 2^5 after 20 years.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q3/5

A quantity triples every year starting at 7. After t years it is

A. $7 + 3t$

B. $21t$

C. $3 \cdot 7^t$

D. $7 \cdot 3^t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q4/5

How many doublings happen in 15 years if the doubling time is 5 years?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • HSA-CED.A.1

BACKWARD 3

Solve linear equations after clearing a denominator • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 500 + 4x = 9x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q1/5

Solve $2 = 0.08(10 + x)$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q2/5

True or false?

To solve $(500 + 4x)/x = 9$ we may multiply both sides by x when $x \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q3/5

2025 = 22.5d gives d =

A. 2002.5

B. 45

C. 9

D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q4/5

Solve $5x/12 = 1/2.4$ for x.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 26 • BACKWARD 3 • Q5/5

SET 26 • HSA-CED.A.1

FORWARD 1

Create equations whose algebraic solutions must be checked against the context • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

A boat travels 24 km downstream and 24 km back in 5 hours. The current is 2 km/h. Write an equation for the boat's still-water speed b and solve it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q1/5

The length of a rectangle is 3 more than its width w, and its diagonal is 15. Write an equation and find w.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q2/5

True or false?

A negative solution of an equation for a time must be rejected.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q3/5

A model gives $t = -3$ or $t = 4$ for the time (in seconds) of an event. The answer is

A. neither

B. both

C. 4 seconds

D. -3 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S26-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q4/5

A pipe fills a tank in x hours and a second pipe takes 2 hours longer. Together they take 2.4 hours. Write an equation and find x.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 26 • FORWARD 1 • Q5/5

SET 26 • HSA-CED.A.1

FORWARD 2

Solve exponential context equations with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

The town of 1,200 doubles every 5 years. Write and solve an equation for when the population reaches 5,000. Give t to the nearest tenth of a year. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q1/5

Solve $500(1.04)^t = 800$ for t. Give an exact answer and a decimal to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q2/5

True or false?

$2^{t/5} = 8$ can be solved without logarithms, but

$2^{t/5} = 5$ cannot be solved exactly without them.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q3/5

The exact solution of $3^x = 20$ is

A. 20/3

B. log(20/3)

C. log 20/log 3

D. 3 log 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q4/5

Explain why log(1.6)/log(1.04) is the time for an amount growing 4% per year to grow by 60%.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 26 • FORWARD 2 • Q5/5

SET 27 • HSA-CED.A.1 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Create and solve nonlinear inequalities from contexts

MAIN	Algebra II main question	**HSA-CED.A.1**
BACKWARD 1	Solve quadratic equations by factoring	**HSA-REI.B.4.b**
BACKWARD 2	Sign of a product on intervals	**7.NS.A.2**
BACKWARD 3	Reverse the inequality when multiplying by a negative	**HSA-REI.B.3**
FORWARD 1	Solve rational inequalities with sign charts and excluded values	**HSA-CED.A.1**
FORWARD 2	Write systems of constraints and judge viability	**HSA-CED.A.3**

SET 27 • HSA-CED.A.1

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

A ball's height is $h = -16t^2 + 64t$ feet after t seconds. Write and solve an inequality for the times when the ball is at least 48 feet high.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q1/5

A company sells x items at a price of 100 − 2x dollars each. Write and solve an inequality for the values of x that give revenue of at least $1,200.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q2/5

The average cost of making x posters is (300 + 2x)/x dollars. Write and solve an inequality for the numbers of posters (x > 0) with an average cost of at most $5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q3/5

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after t hours. Write and solve an inequality for the times when C is at least 4 mg/L.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q4/5

$500 is invested at 4% interest compounded yearly. Write an inequality for the number of whole years t until the value is at least $800, and solve it. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • MAIN • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 1

Solve quadratic equations by factoring • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $t^2 - 4t + 3 = 0$.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q1/5

Solve $x^2 - 50x + 600 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q2/5

True or false?

The solutions of $t^2 - 5t + 4 = 0$ are -1 and -4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q3/5

The solutions of $x^2 - x - 12 = 0$ are

A. 4 and −3

B. 6 and −2

C. 12 and −1

D. −4 and 3

Solve $2x^2 - 100x + 1200 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 2

Sign of a product on intervals • Grade 7 • The Number System • 7.NS.A.2

Is $(t - 1)(t - 3)$ positive or negative when $t = 2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q1/5

Is $(t - 1)(t - 3)$ positive or negative when $t = 5$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q2/5

True or false?

A product of two negative numbers is negative.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q3/5

$(x - 20)(x - 30) \leq 0$ when

A. $x \leq 20$ or $x \geq 30$

B. $x \leq 20$

C. $20 \leq x \leq 30$

D. $x \geq 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q4/5

For which t is $(t - 1)(t - 4) < 0$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.2 • Set 27 • BACKWARD 2 • Q5/5

SET 27 • HSA-CED.A.1

BACKWARD 3

Reverse the inequality when multiplying by a negative • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve $-16t^2 + 64t - 48 \geq 0$ after dividing both sides by -16.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q1/5

Solve $-3x \geq 12$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q2/5

True or false?

Multiplying both sides of an inequality by x is always allowed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q3/5

If $x > 0$, then $(300 + 2x)/x \leq 5$ is equivalent to

A. $300 + 2x \leq 5x$

B. $300 + 2x \geq 5x$

C. $2x \leq 5$

D. $300 \leq 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q4/5

Solve $2 - x < 7$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • HSA-CED.A.1

FORWARD 1

Solve rational inequalities with sign charts and excluded values • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1

Solve $(x - 3)/(x + 1) \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q1/5

Solve $(x + 2)/(x - 4) > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q2/5

True or false?

Multiplying $(x - 3)/(x + 1) \leq 0$ by $x + 1$ gives $x - 3 \leq 0$ for all x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q3/5

The solution set of $1/x > 2$ is

A. $x > 1/2$

B. $x < 1/2$

C. $x > 0$

D. $0 < x < 1/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q4/5

A model's average cost is (400 + 3x)/x dollars. For how many items x > 0 is the average cost under $7?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.1 • Set 27 • FORWARD 1 • Q5/5

SET 27 • HSA-CED.A.1

FORWARD 2

Write systems of constraints and judge viability • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

The ball's height must be at least 48 ft and the time at most 2.5 s. Write the constraints and find the viable times.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q1/5

Revenue $x(100 - 2x)$ must be at least \$1,200 and at most 24 items can be made. Which x are viable?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q2/5

True or false?

A value that solves the inequality may still not be viable in the context.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q3/5

Revenue $x(100 - 2x) \geq 1200$ with x a whole number is satisfied by

A. $x = 19$

B. $x = 25$

C. $x = 31$

D. $x = 50$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q4/5

Explain why 12 years, not 11.98 years, answers the compound-interest question.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 27 • FORWARD 2 • Q5/5

Create and graph nonlinear two-variable models

MAIN	Algebra II main question	**HSA-CED.A.2**
BACKWARD 1	Make a table of values from an equation	**6.EE.C.9**
BACKWARD 2	Choose scales and label axes	**HSN-Q.A.1**
BACKWARD 3	Write a growth or decay model from a percent	**HSF-LE.A.2**
FORWARD 1	Create equations in more than two variables and graph a member	**HSA-CED.A.2**
FORWARD 2	Interpret key features of the created models	**HSF-IF.B.4**

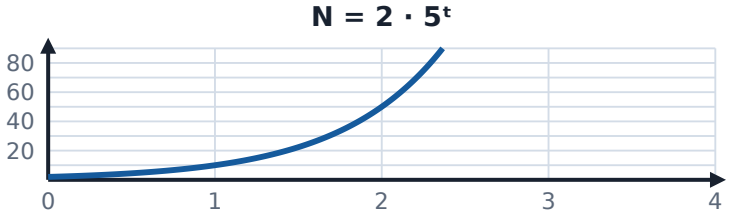
SET 28 • HSA-CED.A.2

MAIN

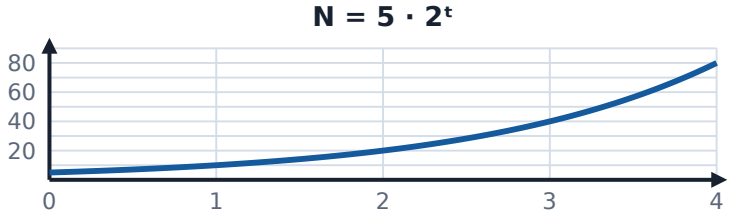
Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

A rabbit population starts at 5 and doubles every year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 4$ (t in years, N rabbits)?

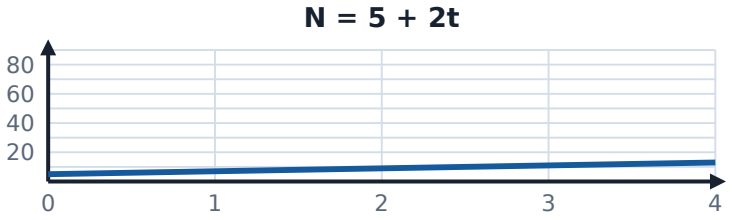
A



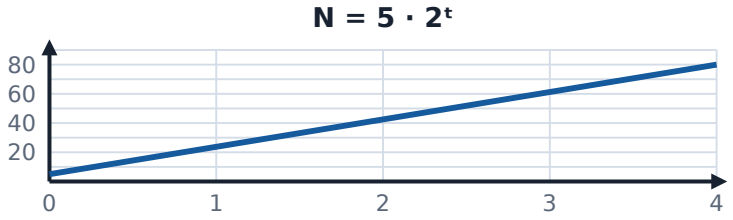
B



C



D



A. Graph A

B. Graph B

C. Graph C

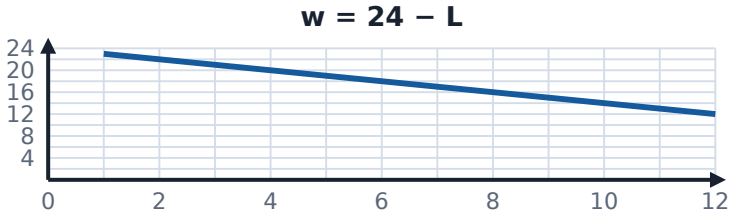
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q1**

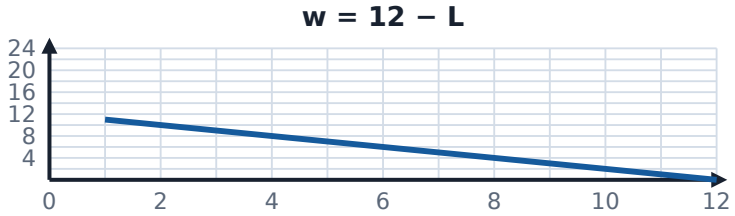
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q1/5

A rectangle has area 24 square units. Which graph shows a correct equation for the situation with a correct graph of that equation for $1 \le L \le 12$ (length L, width w)?

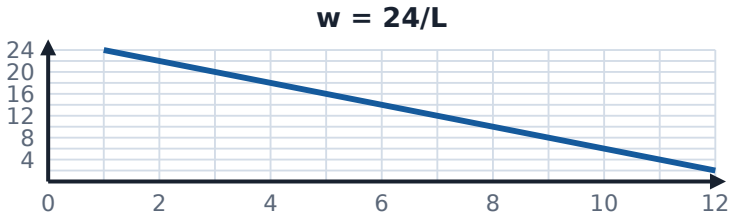
A



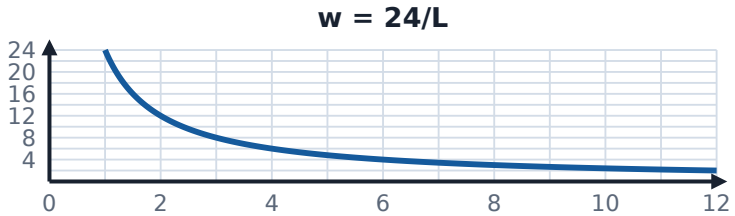
B



C



D



A. Graph A

B. Graph B

C. Graph C

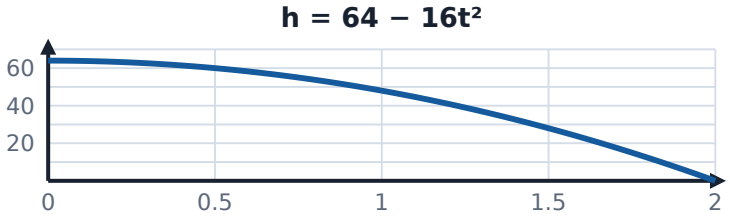
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q2**

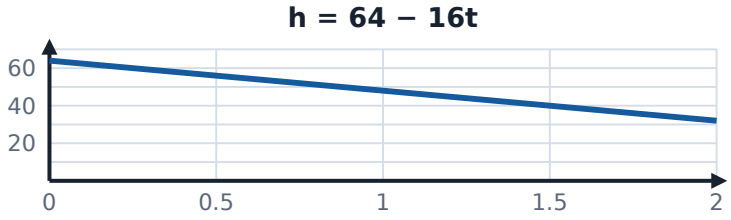
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q2/5

A stone is dropped from 64 feet. Its height after t seconds is 64 minus 16 times the square of t. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 2$ (h in feet)?

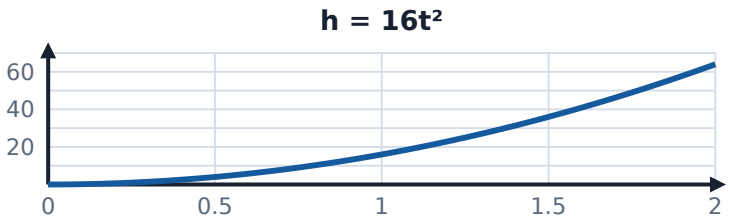
A



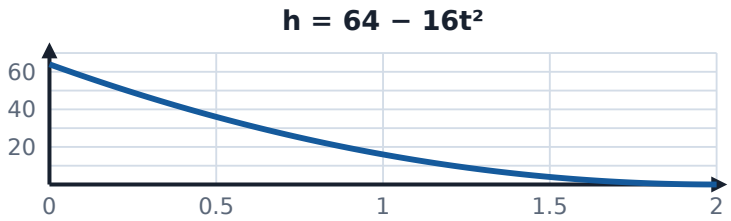
B



C



D



A. Graph A

B. Graph B

C. Graph C

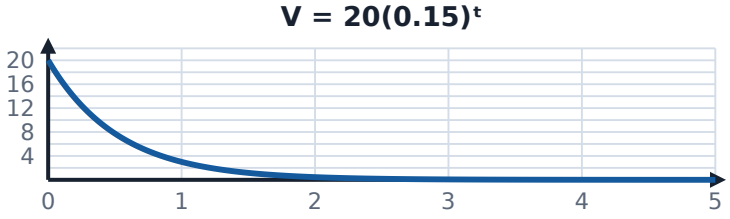
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q3**

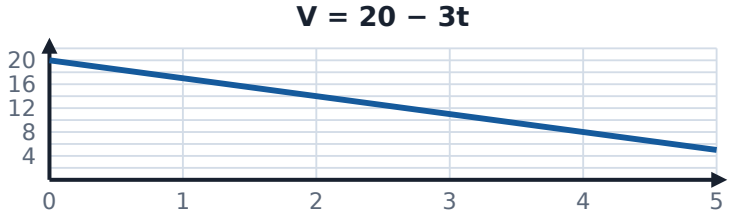
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q3/5

A $20,000 car loses 15% of its value each year. Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \le t \le 5$ (V in thousands of dollars, t in years)?

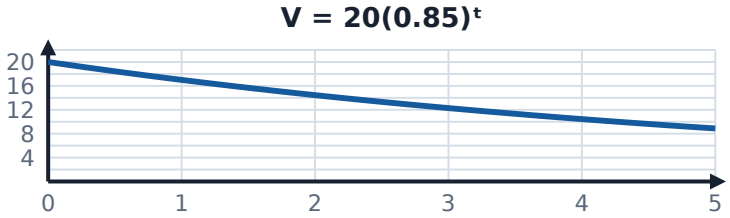
A



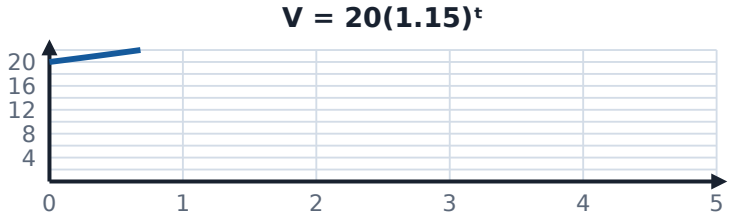
B



C



D



A. Graph A

B. Graph B

C. Graph C

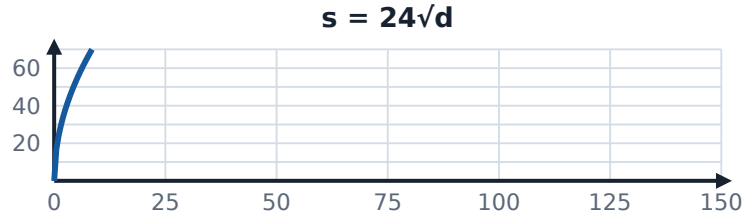
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q4**

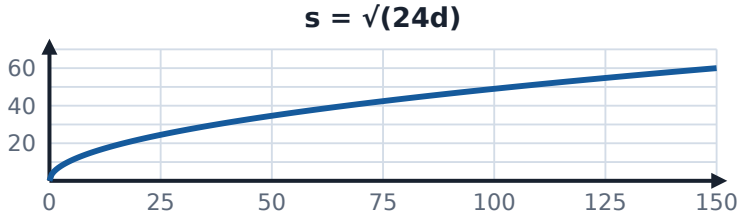
High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q4/5

A car's speed s (mph) is the square root of 24 times its skid length d (ft). Which graph shows a correct equation for the situation with a correct graph of that equation for $0 \leq d \leq 150$?

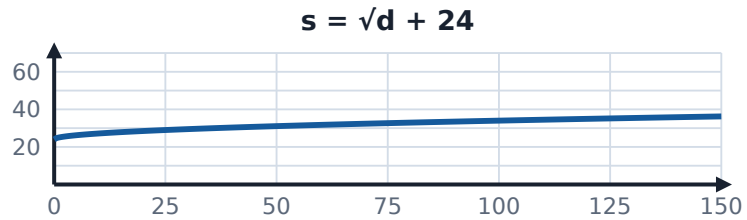
A



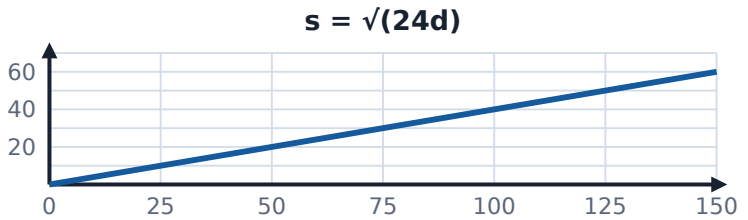
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-M-Q5**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • MAIN • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 1

Make a table of values from an equation • Grade 6 • Expressions & Equations • 6.EE.C.9

For $N = 200 \cdot 2^t$, find N when t = 0, 1, 2, 3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

For w = 24/L, find w when L = 2, 3, 4, 6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

True or false?

For $h = 64 - 16t^2$, $h = 48$ when $t = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

For $s = \sqrt{(24d)}$, s when $d = 96$ is

A. 96

B. 48

C. 2304

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

For $V = 20000(0.85)^t$, find V when t = 1 and t = 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 2

Choose scales and label axes • High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1

Values of N go from 200 to 3,200. Give a sensible step for the vertical axis on a grid with about 7 lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q1/5

Which quantity goes on the horizontal axis when V depends on t?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q2/5

True or false?

Axis labels should include units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q3/5

The best vertical scale for values from 0 to 64 is

A. steps of 0.5

B. steps of 1 from 0 to 10

C. steps of 100 from 0 to 1000

D. steps of 10 from 0 to 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q4/5

Give the horizontal range for $0 \leq d \leq 150$ with a step of 10. How many steps is that?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

High School — Algebra I • Number & Quantity: Quantities • HSN-Q.A.1 • Set 28 • BACKWARD 2 • Q5/5

SET 28 • HSA-CED.A.2

BACKWARD 3

Write a growth or decay model from a percent • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the value of $20,000 after losing 15% per year for t years.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q1/5

What is the growth factor for 4% growth?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q2/5

True or false?

Losing 15% each year means multiplying by 0.15 each year.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q3/5

A quantity doubles every hour starting at 200. After t hours it is

A. $2 \cdot 200^{t}$

B. $200 \cdot 2^{t}$

C. $200 + 2t$

D. $400t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q4/5

What is the decay factor for 30% decay?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 28 • BACKWARD 3 • Q5/5

SET 28 • HSA-CED.A.2

FORWARD 1

Create equations in more than two variables and graph a member of the family • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2

The current I (amps) through a resistor is the voltage V divided by the resistance R. Write the equation, and write I as a function of R when V = 12.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q1/5

The volume of a cylinder is $V = \pi r^2 h$. Write h in terms of r when $V = 100\pi$, and describe its graph for $r > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q2/5

True or false?

Fixing one variable of a three-variable equation gives a two-variable equation that can be graphed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q3/5

For $A = P(1.05)^t$ with $P = 1000$, the graph of A against t is

A. a line

B. an increasing exponential curve

C. a parabola

D. a decreasing curve

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q4/5

For the area formula $A = \pi r^2$, which graph (A against r) is correct for $r \geq 0$: a line, a half parabola, or an exponential?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.2 • Set 28 • FORWARD 1 • Q5/5

SET 28 • HSA-CED.A.2

FORWARD 2

Interpret key features of the created models • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

For $N = 200 \cdot 2^t$, what does the N-intercept mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q1/5

For $h = 64 - 16t^2$, when does the stone hit the ground, and what is the t-intercept's meaning?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q2/5

True or false?

The graph of $w = 24/L$ never reaches the L-axis because a rectangle of area 24 cannot have width 0.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q3/5

For $V = 20000(0.85)^t$, the graph

A. decreases and approaches 0

B. increases

C. is a line

D. decreases to 0 at t = 6.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q4/5

Describe the rate of change of $s = \sqrt{(24d)}$ as d grows.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 28 • FORWARD 2 • Q5/5

Represent nonlinear constraints and judge viable options

MAIN	Algebra II main question	**HSA-CED.A.3**
BACKWARD 1	Represent constraints with a linear system	**HSA-CED.A.3**
BACKWARD 2	Test whether a point satisfies an inequality	**6.EE.B.5**
BACKWARD 3	Solve quadratic inequalities	**HSA-CED.A.1**
FORWARD 1	Optimize within nonlinear constraints	**HSA-CED.A.3**
FORWARD 2	Locate constraint boundaries as graph intersections	**HSA-REI.D.11**

SET 29 • HSA-CED.A.3

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

A garden has length L and width W (feet). The fencing for L + W is 30 ft and the area must be at least 200 ft². Write the constraints, find the viable widths, and decide whether W = 8 is viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q1/5

A box has a square base of side x and a volume of 32 in^3; both x and the height h must be at most 4 in. Write the constraints, find the viable x, and decide whether x = 3 is viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q2/5

A plan invests $1,000 at 5% compounded yearly and needs at least $1,500 within 10 years. Write the constraint and decide whether the plan is viable. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q3/5

Attendance at a show is $500 - 10p$ people when tickets cost p dollars. Revenue must be at least $6,000. Write the constraints, find the viable prices, and decide whether $35 is viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q4/5

A rectangular pen with sides x and y (m) needs area at least 50 m² and perimeter at most 40 m. Write the system and decide whether (5, 10) and (4, 12) are viable.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • MAIN • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 1

Represent constraints with a linear system • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3

Write a constraint for "the total of L and W is 30".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q1/5

Write a constraint for "at most 40 m of fence for a rectangle with sides x and y".

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q2/5

True or false?

"At least 200" is written ≤ 200.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q3/5

"The height h must be at most 4" is

A. $h \leq 4$

B. $h \geq 4$

C. $h < 4$

D. $h = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q4/5

Is (12, 20) a solution of $x + y \leq 30$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 2

Test whether a point satisfies an inequality • Grade 6 • Expressions & Equations • 6.EE.B.5

Does $W = 8$ satisfy $W(30 - W) \geq 200$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q1/5

Does $p = 25$ satisfy $p(500 - 10p) \geq 6000$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q2/5

True or false?

(4, 12) satisfies $xy \geq 50$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q3/5

Which x satisfies $x^2 \geq 8$?

A. 3

B. 2

C. 1

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q4/5

Does $t = 10$ satisfy $1000(1.05)^t \geq 1500$? A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • HSA-CED.A.3

BACKWARD 3

Solve quadratic inequalities • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1

Solve $W^2 - 30W + 200 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q1/5

Solve $p^2 - 50p + 600 \leq 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q2/5

True or false?

$x^2 \geq 8$ with $x > 0$ means $x \geq 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q3/5

The solution of $(x - 2)(x - 6) \leq 0$ is

A. $x = 2$ or $x = 6$

B. $2 \leq x \leq 6$

C. $x \leq 2$ or $x \geq 6$

D. $x \geq 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q4/5

Solve $32/x^2 \leq 4$ for $x > 0$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.1 • Set 29 • BACKWARD 3 • Q5/5

SET 29 • HSA-CED.A.3

FORWARD 1

Optimize within nonlinear constraints • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3

A rectangular garden has length L and width W (feet) with L + W = 30. Which width gives the greatest area, and what is that area?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q1/5

Which ticket price p gives the greatest revenue $p(500 - 10p)$, and what is that revenue?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q2/5

True or false?

A maximum found from the model must still satisfy every other constraint to be a viable choice.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q3/5

For the pen with $xy \geq 50$ and $2x + 2y \leq 40$, which is viable?

A. (10, 10)

B. (15, 10)

C. (2, 20)

D. (4, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q4/5

Among whole-number widths, list all viable garden widths when the area must be at least 216 ft².

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.3 • Set 29 • FORWARD 1 • Q5/5

SET 29 • HSA-CED.A.3

FORWARD 2

Locate constraint boundaries as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Explain how the viable widths $10 \leq W \leq 20$ can be read from the graphs of $y = W(30 - W)$ and $y = 200$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q1/5

Where do $y = p(500 - 10p)$ and $y = 6000$ intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q2/5

True or false?

The x-coordinates where $y = 32/x^2$ meets $y = 4$ solve $32/x^2 = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q3/5

The graphs of $y = 1000(1.05)^t$ and $y = 1500$ meet at $t \approx$

A. 15

B. 8.3

C. 10

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S29-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q4/5

Explain why the region above y = 200 and below the parabola shows the viable widths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 29 • FORWARD 2 • Q5/5

Rearrange rational, radical, and exponential formulas

MAIN	Algebra II main question	**HSA-CED.A.4**
BACKWARD 1	Rearrange linear formulas	**HSA-CED.A.4**
BACKWARD 2	Undo squares and square roots	**8.EE.A.2**
BACKWARD 3	Clear fractions in an equation	**HSA-REI.B.3**
FORWARD 1	Rearrange formulas where conditions decide the result	**HSA-CED.A.4**
FORWARD 2	Rearrange exponential formulas with logarithms	**HSF-LE.A.4**

SET 30 • HSA-CED.A.4

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

The lens formula is $1/f = 1/a + 1/b$. Solve for b and state the condition needed.

$b =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q1/5

A pendulum's period is $T = 2\pi\sqrt{(L/g)}$. Solve for L.

$L =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q2/5

An investment grows by $A = P(1 + r)^t$. Solve for r.

r = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q3/5

The speed of a falling object is $v = \sqrt{(2gh)}$. Solve for h and state the condition on v.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q4/5

In a circuit, $I = V/(R + r)$. Solve for R and state the condition needed.

$R =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • MAIN • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 1

Rearrange linear formulas • High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4

Solve P = 2L + 2W for W.

W = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q1/5

Solve d = rt for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q2/5

True or false?

Solving V = IR for R gives R = VI.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q3/5

Solving $y = mx + b$ for x gives

A. $(y - b)/m$

B. $y/m - b$

C. $(y + b)/m$

D. $m(y - b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q4/5

Solve A = (1/2)bh for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

High School — Algebra I • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 2

Undo squares and square roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $x^2 = 49$ for $x > 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q1/5

Solve $\sqrt{y} = 6$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q2/5

True or false?

If $\sqrt{(L/g)} = k$, then $L/g = \sqrt{k}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q3/5

$(\sqrt{(2gh)})^2$ equals

A. $2g\sqrt{h}$

B. $2gh$

C. $\sqrt{(2gh)}$

D. $4g^2h^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q4/5

Solve $T/(2\pi) = \sqrt{m}$ for m.

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • HSA-CED.A.4

BACKWARD 3

Clear fractions in an equation • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 1/2 + 1/x = 1 for x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q1/5

Combine 1/a − 1/f into one fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q2/5

True or false?

If $1/b = (a - f)/(af)$, then $b = af/(a - f)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q3/5

Solving $3 = 12/(R + 2)$ for R gives

A. 36

B. 4

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q4/5

Solve $I = V/(R + r)$ for $R + r$.

$R + r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 30 • BACKWARD 3 • Q5/5

SET 30 • HSA-CED.A.4

FORWARD 1

Rearrange formulas where conditions decide the result • High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4

Solve $A = \pi r^2$ for r, given r is a length.

$r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q1/5

Solve $y = (x + 1)/(x - 2)$ for x and state the condition on y.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q2/5

True or false?

Solving $s = \sqrt{(24d)}$ for d gives $d = s^2/24$ for every real s.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q3/5

Solving $E = (1/2)mv^2$ for v ($v \geq 0$) gives

A. $\sqrt{(2E/m)}$

B. $(2E/m)^2$

C. $\sqrt{(E/2m)}$

D. $2E/m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q4/5

Solve $P = k/V^2$ for V ($V > 0$).

$V =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Creating Equations • HSA-CED.A.4 • Set 30 • FORWARD 1 • Q5/5

SET 30 • HSA-CED.A.4

FORWARD 2

Rearrange exponential formulas with logarithms • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $A = Pe^{rt}$ for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q1/5

Solve $A = P(1 + r)^t$ for t.

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q2/5

True or false?

Solving $N = N_0 \cdot 2^{t/d}$ for t gives $t = d \cdot \log_2(N/N_0)$.

A. True

B. False

Solving $y = 5 \cdot 10^x$ for x gives

A. y/50

B. log(y/5)

C. $10^{y/5}$

D. log y − 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q4/5

Use $t = \log(A/P)/\log(1 + r)$ to find t for $A = 2000$, $P = 1000$, $r = 0.05$. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 30 • FORWARD 2 • Q5/5

SET 31 • HSA-REI.A.2 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Solve rational equations and identify extraneous solutions

MAIN	Algebra II main question	**HSA-REI.A.2**
BACKWARD 1	Solve proportions	**7.RP.A.2**
BACKWARD 2	Find a common denominator of rational expressions	**HSA-APR.D.7**
BACKWARD 3	Values that make a denominator zero	**7.NS.A.2.b**
FORWARD 1	Explain and construct extraneous solutions	**HSA-REI.A.2**
FORWARD 2	See rational equations as graph intersections	**HSA-REI.D.11**

SET 31 • HSA-REI.A.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $3/(x - 1) = 6/(x + 2)$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q1/5

Solve $x/(x - 2) + 1 = 2/(x - 2)$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q2/5

Solve 1/x + 1/(x + 2) = 4/3. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q3/5

Solve $x^2/(x - 3) = 9/(x - 3)$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q4/5

Solve $2/(x + 1) - 1/(x - 1) = 2/(x^2 - 1)$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • MAIN • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 1

Solve proportions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2

Solve $3/x = 6/10$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q1/5

Solve 4/7 = x/21.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q2/5

True or false?

$a/b = c/d$ implies $ac = bd$ when $b, d \neq 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q3/5

2/5 = 6/x gives x =

A. 12

B. 2.4

C. 3

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q4/5

Solve $3/(x - 1) = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2 • Set 31 • BACKWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 2

Find a common denominator of rational expressions • High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.D.7

What is the LCD of 1/x and 1/(x + 2)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q1/5

What is the LCD of $\frac{2}{x+1}$, $\frac{1}{x-1}$, and $\frac{2}{x^2-1}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q2/5

True or false?

The LCD of $1/(x-2)$ and $2/(x-2)$ is $(x-2)^2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q3/5

Multiplying $1/x + 1/(x + 2) = 4/3$ by $3x(x + 2)$ gives

A. $3 + 3 = 4$

B. $3(x + 2) + 3x = 4$

C. $3(x + 2) + 3x = 4x(x + 2)$

D. $x + 2 + x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q4/5

Write 1/x + 1/(x + 2) as one fraction.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

High School — Algebra II, (+) standard listed in the pathway • Algebra: Arithmetic with Polynomials & Rational Expressions
HSA-APR.D.7 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • HSA-REI.A.2

BACKWARD 3

Values that make a denominator zero • Grade 7 • The Number System • 7.NS.A.2.b

Which x-values make $x/(x - 2)$ undefined?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q1/5

Which x-values make $2/(x^2 - 1)$ undefined?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q2/5

True or false?

A rational equation cannot have a solution that makes any denominator zero.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q3/5

Which value is excluded from $1/x + 1/(x + 2) = 4/3$?

A. 2

B. 1

C. 4/3

D. -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q4/5

List the excluded values of $x^2/(x - 3) = 9/(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 31 • BACKWARD 3 • Q5/5

SET 31 • HSA-REI.A.2

FORWARD 1

Explain and construct extraneous solutions • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Explain how multiplying both sides of $x/(x - 2) + 1 = 2/(x - 2)$ by $x - 2$ creates an extraneous solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q1/5

Write a rational equation that has the extraneous solution x = 4 and no actual solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q2/5

True or false?

Every rational equation has at least one extraneous solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q3/5

For which a does $a/(x - 1) = 2/(x - 1)$ have infinitely many solutions?

A. a = 2

B. no value

C. a = 0

D. a = 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q4/5

Solve $(x + 1)/(x - 1) = 2/(x - 1) + 1$ and explain the result.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 31 • FORWARD 1 • Q5/5

SET 31 • HSA-REI.A.2

FORWARD 2

See rational equations as graph intersections • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Explain why $x/(x - 2) + 1 = 2/(x - 2)$ has no solution using the graphs of $y = x/(x - 2) + 1$ and $y = 2/(x - 2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q1/5

Where do $y = 3/(x - 1)$ and $y = 6/(x + 2)$ intersect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q2/5

True or false?

An extraneous solution corresponds to an intersection point of the two graphs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q3/5

$y = x^2/(x - 3)$ and $y = 9/(x - 3)$ intersect at

A. $(3, 3/2)$

B. $(-3, 3/2)$

C. $(-3, -3/2)$

D. $(3, 0)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q4/5

Use a graph or table to approximate the solution of $1/x = x - 1$ with $x > 0$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 31 • FORWARD 2 • Q5/5

SET 32 • HSA-REI.A.2 • CREATING EQUATIONS AND REASONING WITH EQUATIONS

Solve radical equations and identify extraneous solutions

MAIN	Algebra II main question	**HSA-REI.A.2**
BACKWARD 1	Square and cube both sides; principal roots	**8.EE.A.2**
BACKWARD 2	Solve the resulting quadratic	**HSA-REI.B.4.b**
BACKWARD 3	Check a candidate in the original equation	**6.EE.B.5**
FORWARD 1	Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$	**HSA-REI.A.2**
FORWARD 2	Explain extraneous solutions with graphs of radical functions	**HSF-IF.C.7.b**

SET 32 • HSA-REI.A.2

MAIN

Algebra II main question • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 7)} = x - 5$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q1/5

Solve $\sqrt{(2x - 1)} = 3$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q2/5

Solve $\sqrt{(x + 3)} + 3 = x$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q3/5

Solve $\sqrt[3]{(x-2)} = -3$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q4/5

Solve $\sqrt{(x - 1)} + 5 = 2$. Identify any extraneous solutions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • MAIN • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 1

Square and cube both sides; principal roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Solve $\sqrt{y} = 3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q1/5

Solve $\sqrt[3]{y} = -3$.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q2/5

True or false?

$\sqrt{9} = 3$ or -3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q3/5

$\sqrt{y} = -2$ has

A. $y = 2$

B. no solution

C. $y = 4$

D. $y = -4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q4/5

Square both sides of $\sqrt{(x + 7)} = x - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 2

Solve the resulting quadratic • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 - 11x + 18 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q1/5

Solve $x^2 - 7x + 6 = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q2/5

True or false?

$(x - 5)^2 = x^2 + 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q3/5

$x^2 - 6x + 9 = x + 3$ gives

A. $x^2 - 7x + 12 = 0$

B. $x^2 - 5x + 12 = 0$

C. $x^2 - 6x + 6 = 0$

D. $x^2 - 7x + 6 = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q4/5

Solve $x^2 - 4x = 0$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 32 • BACKWARD 2 • Q5/5

SET 32 • HSA-REI.A.2

BACKWARD 3

Check a candidate in the original equation • Grade 6 • Expressions & Equations • 6.EE.B.5

Check $x = 2$ in $\sqrt{(x + 7)} = x - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q1/5

Check $x = 6$ in $\sqrt{(x + 3)} + 3 = x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q2/5

True or false?

$x = 4$ satisfies $\sqrt{(2x - 1)} = 3$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q3/5

Which value satisfies $\sqrt{(x + 3)} + 3 = x$?

A. 1

B. 0

C. 6

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q4/5

Check $x = 10$ in $\sqrt{(x - 1)} + 5 = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 32 • BACKWARD 3 • Q5/5

SET 32 • HSA-REI.A.2

FORWARD 1

Solve equations with two radicals, rational exponents, and $\sqrt{(x^2)}$ • High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2

Solve $\sqrt{(x + 5)} - \sqrt{x} = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q1/5

Solve $x^{3/2} = 8$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q2/5

Solve $\sqrt{(x^2)} = 5$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q3/5

True or false?

$\sqrt{(x^2)} = x$ for every real x.

A. True

B. False

The solution set of $\sqrt{(x^2)} = x$ is

A. $x \geq 0$

B. $x \leq 0$

C. $x = 0$ only

D. all real x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

High School — Algebra II • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.2 • Set 32 • FORWARD 1 • Q5/5

SET 32 • HSA-REI.A.2

FORWARD 2

Explain extraneous solutions with graphs of radical functions • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b

The graphs of $y = \sqrt{(x + 7)}$ and $y = x - 5$ meet at one point. Give it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q1/5

The extraneous value $x = 2$ is where $y = x - 5$ meets which other graph?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q2/5

True or false?

The graph of $y = \sqrt{(x + 7)}$ starts at $(7, 0)$ and increases.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q3/5

The graph of $y = \sqrt{(x - 1)} + 5$ lies

A. below $y = 2$

B. below the x-axis

C. at or below $y = 5$

D. at or above $y = 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S32-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q4/5

Use the graphs to explain why $\sqrt{(x - 1)} + 5 = 2$ has no solution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 32 • FORWARD 2 • Q5/5

Explain why intersections of nonlinear graphs solve f(x) = g(x)

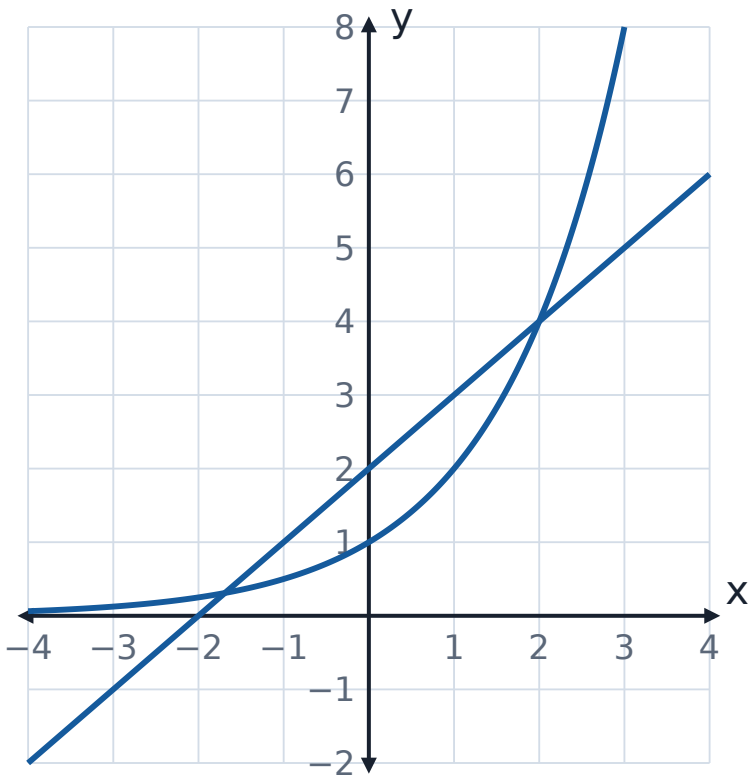
MAIN	Algebra II main question	**HSA-REI.D.11**
BACKWARD 1	A graph is the set of its solutions	**HSA-REI.D.10**
BACKWARD 2	Solve a linear system by graphing	**8.EE.C.8.a**
BACKWARD 3	Evaluate nonlinear functions at an input	**HSF-IF.A.2**
FORWARD 1	Relate intersections to zeros of f - g	**HSA-REI.D.11**
FORWARD 2	Find all solutions of a trigonometric equation on an interval from a…	**HSF-TF.B.7**

SET 33 • HSA-REI.D.11

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

The graphs of $y = 2^x$ and $y = x + 2$ are shown. They meet at $x = 2$ and at a negative x. Explain why $x = 2$ solves $2^x = x + 2$, and verify it.



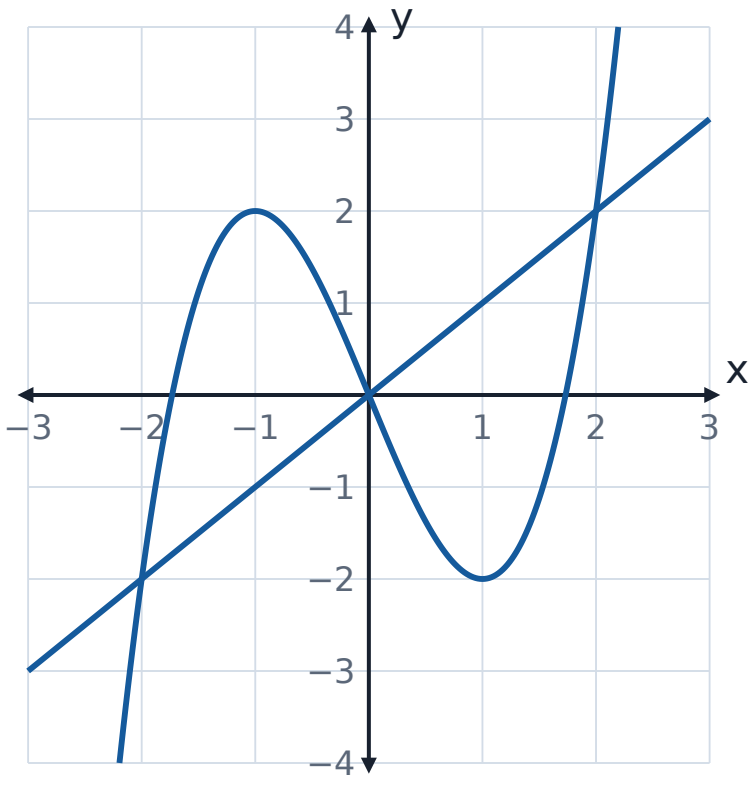
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q1/5

The graphs of $y = x^3 - 3x$ and $y = x$ are shown. Use them to list every solution of $x^3 - 3x = x$, and explain why they are solutions.



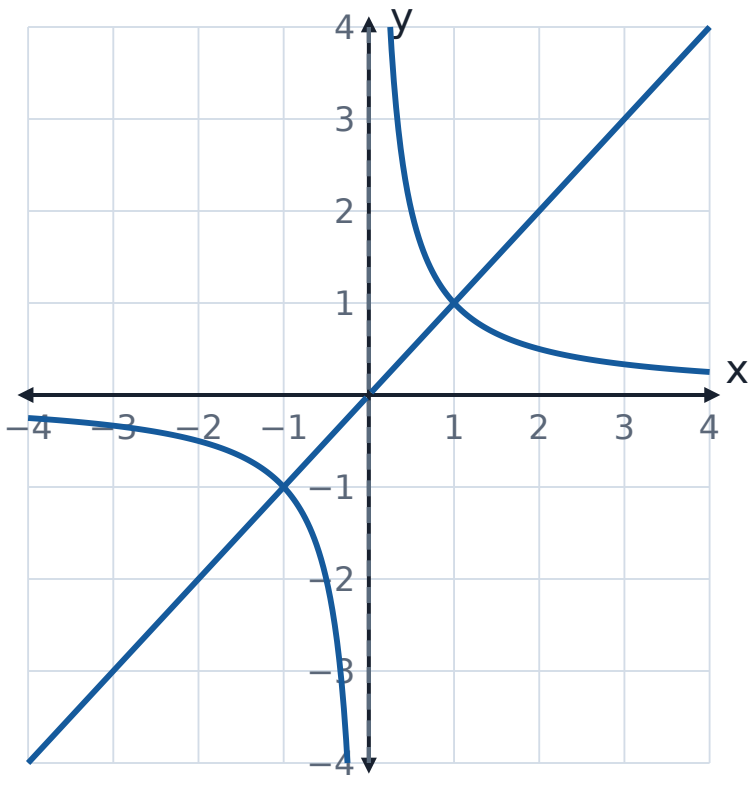
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q2/5

The graphs of $y = 1/x$ and $y = x$ are shown. List every solution of $1/x = x$ and explain using the graphs.



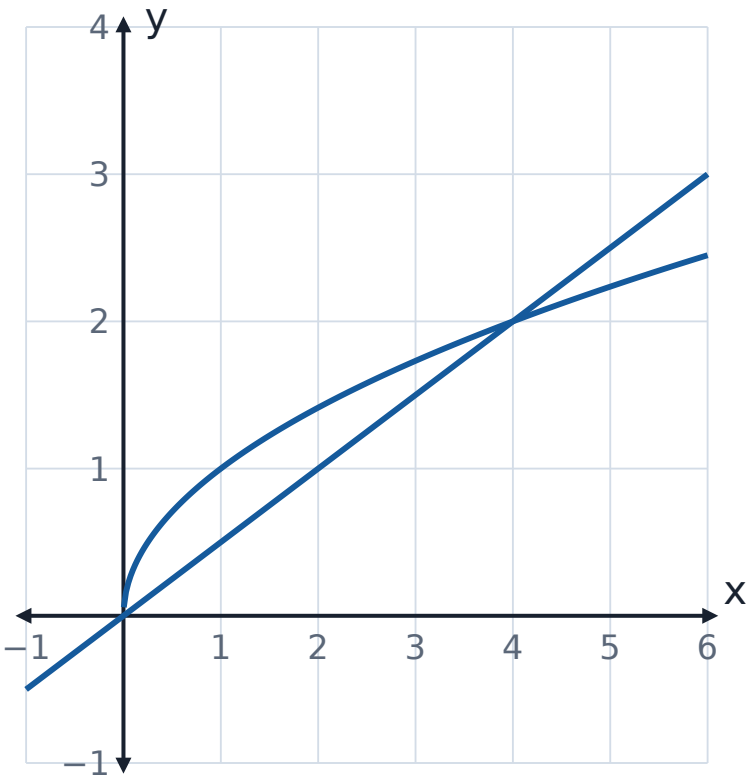
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q3/5

The graphs of $y = \sqrt{x}$ and $y = x/2$ are shown. List every solution of $\sqrt{x} = x/2$ and verify each.



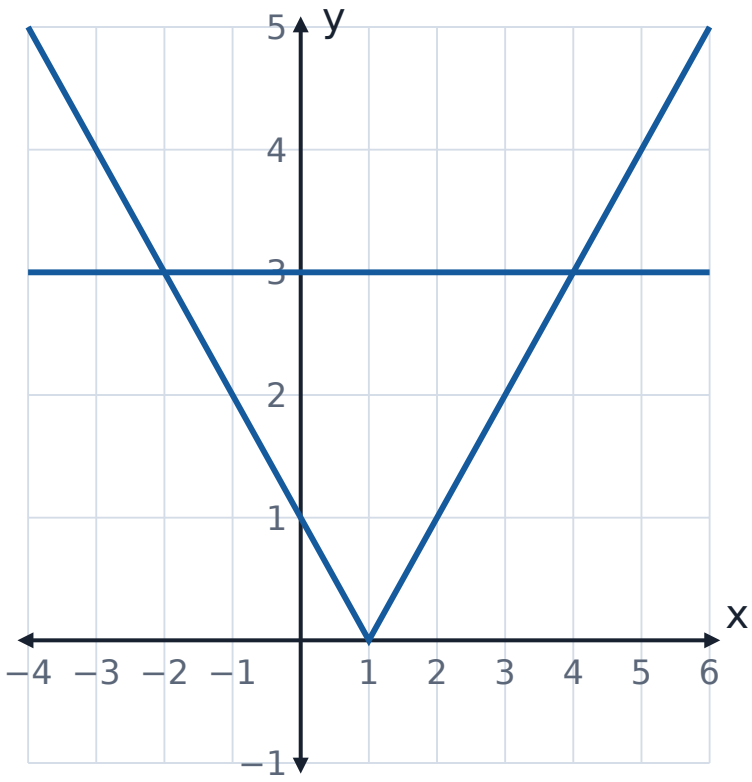
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q4/5

The graphs of $y = |x - 1|$ and $y = 3$ are shown. List every solution of $|x - 1| = 3$ and explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • MAIN • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 1

A graph is the set of its solutions • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10

Is (2, 4) on the graph of $y = 2^x$? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q1/5

Is (3, 6) on the graph of $y = x + 2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q2/5

True or false?

Every point on the graph of y = f(x) has the form (x, f(x)).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q3/5

Which point is on $y = x^3 - 3x$?

A. (0, 3)

B. (2, 6)

C. (2, 2)

D. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q4/5

Is $(-1, -1)$ on the graph of $y = 1/x$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.10 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 2

Solve a linear system by graphing • Grade 8 • Expressions & Equations • 8.EE.C.8.a

The lines $y = x + 1$ and $y = -x + 5$ meet at $(2, 3)$. Explain why $x = 2$ solves $x + 1 = -x + 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q1/5

Where do y = 2x and y = 6 meet?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q2/5

True or false?

Two lines that do not intersect give an equation with no solution.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q3/5

$y = 3$ and $y = x - 1$ meet at $x =$

A. 4

B. 3

C. -1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q4/5

Solve $x + 1 = -x + 5$ algebraically.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 33 • BACKWARD 2 • Q5/5

SET 33 • HSA-REI.D.11

BACKWARD 3

Evaluate nonlinear functions at an input • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find 2^x and $x + 2$ when $x = 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q1/5

Find $x^3 - 3x$ when $x = -2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q2/5

True or false?

$\sqrt{9} = 9/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q3/5

$|x - 1|$ when $x = -2$ is

A. 3

B. 1

C. -1

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q4/5

Find $1/x$ when $x = -1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 33 • BACKWARD 3 • Q5/5

SET 33 • HSA-REI.D.11

FORWARD 1

Relate intersections to zeros of f − g • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Explain why the solutions of $x^3 - 3x = x$ are the zeros of $h(x) = x^3 - 4x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q1/5

Find the zeros of $h(x) = x^3 - 4x$ algebraically.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q2/5

True or false?

If $y = f(x)$ and $y = g(x)$ meet at $x = a$, then the graph of $y = f(x) - g(x)$ crosses or touches the x-axis at a.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q3/5

The solutions of $\sqrt{x} = x/2$ are the zeros of

A. $\sqrt{x} + x/2$

B. $\sqrt{x} - x/2$

C. $\sqrt{x} \cdot x/2$

D. $x/2$

Explain why $2^x = x + 2$ has exactly two solutions using $h(x) = 2^x - x - 2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 33 • FORWARD 1 • Q5/5

SET 33 • HSA-REI.D.11

FORWARD 2

Find all solutions of a trigonometric equation on an interval from a graph • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

On $0 \le x < 2\pi$, where does $y = \sin x$ meet $y = 1/2$? Give exact values.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q1/5

On $0 \le x < 2\pi$, where does $y = \cos x$ meet $y = 0$?

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q2/5

True or false?

sin x = 1/2 has exactly two real solutions.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q3/5

How many solutions does $\sin x = 0.3$ have on $0 \leq x < 2\pi$?

A. 1

B. 0

C. 2

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S33-F2-Q4**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q4/5

Explain why $\sin x = 2$ has no solution using the graph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 33 • FORWARD 2 • Q5/5

Approximate solutions of f(x) = g(x)

MAIN	Algebra II main question	**HSA-REI.D.11**
BACKWARD 1	Read solutions of linear-quadratic systems from graphs	**HSA-REI.D.11**
BACKWARD 2	Evaluate exponential and logarithmic expressions with a calculat…	**HSF-IF.A.2**
BACKWARD 3	Round to a place value	**5.NBT.A.4**
FORWARD 1	Guarantee the accuracy of an approximation	**HSA-REI.D.11**
FORWARD 2	Compare an approximation with an exact logarithmic solution	**HSF-LE.A.4**

SET 34 • HSA-REI.D.11

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Use a table or successive approximation to solve $2^x = 5 - x$ to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q1/5

Solve $x^3 = 7 - x$ to the nearest hundredth with a table or successive approximation. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q2/5

Solve $\log x = 2 - x$ (log base 10) to the nearest hundredth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q3/5

The table shows $f(x) = 3(0.8)^x$ and $g(x) = x$. Between which two table values is the solution of $f(x) = g(x)$? Estimate it to the nearest tenth.

x	0	0.5	1	1.5	2	2.5
f(x)	3	2.683	2.4	2.147	1.92	1.717
g(x)	0	0.5	1	1.5	2	2.5

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q4/5

Use a graph or table to find the negative solution of $2^x = x + 2$ to the nearest tenth. A calculator may be used.

$x \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • MAIN • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 1

Read solutions of linear-quadratic systems from graphs • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

Where do $y = x^2$ and $y = 4$ meet?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q1/5

Read the intersection of $y = x^2$ and $y = x + 2$ from a graph or table.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q2/5

True or false?

The solutions of $f(x) = g(x)$ can be approximated from a table where $f - g$ changes sign.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q3/5

$f(1) - g(1) = -2$ and $f(2) - g(2) = 1$. A solution lies

A. below 1

B. at 1.5 exactly

C. above 2

D. between 1 and 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q4/5

$f(x) = x^2$ and $g(x) = 3$. Between which consecutive integers is the positive solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 2

Evaluate exponential and logarithmic expressions with a calculator • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

Find $2^{1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q1/5

Find log 1.75 (base 10) to the nearest thousandth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q2/5

True or false?

$1.7^3 = 5.1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q3/5

$3(0.8)^2$ equals

A. 0.192

B. 4.8

C. 1.92

D. 2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q4/5

Find $2^{-1.7}$ to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • HSA-REI.D.11

BACKWARD 3

Round to a place value • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round 1.7156 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q1/5

Round 1.944 to the nearest tenth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q2/5

True or false?

1.755 rounded to the nearest hundredth is 1.76.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q3/5

-1.690 to the nearest tenth is

A. -1.6

B. -1.69

C. -2

D. -1.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q4/5

Round 1.7392 to the nearest hundredth.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 34 • BACKWARD 3 • Q5/5

SET 34 • HSA-REI.D.11

FORWARD 1

Guarantee the accuracy of an approximation • High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11

$h(x) = 2^x + x - 5$. Find $h(1.71)$ and $h(1.72)$ and explain why a solution lies between them. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q1/5

Use one step of bisection on $[1, 2]$ for $x^3 + x - 7 = 0$. Which half contains the solution?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q2/5

True or false?

After 4 bisection steps starting from an interval of length 1, the interval has length 1/8.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q3/5

How many bisection steps from [1, 2] guarantee an interval shorter than 0.01?

A. 5

B. 100

C. 10

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q4/5

Explain why checking h(1.715) and h(1.725) decides whether the solution rounds to 1.72.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Algebra: Reasoning with Equations & Inequalities • HSA-REI.D.11 • Set 34 • FORWARD 1 • Q5/5

SET 34 • HSA-REI.D.11

FORWARD 2

Compare an approximation with an exact logarithmic solution • High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4

Solve $2^x = 5$ exactly, then give a decimal to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q1/5

Solve $3(0.8)^x = 1$ exactly and to the nearest hundredth. A calculator may be used.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q2/5

True or false?

$2^x = 5 - x$ can be solved exactly with one logarithm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q3/5

Which equation can be solved exactly with logarithms?

A. $\log x = 2 - x$

B. $2^x = x + 2$

C. $5(1.2)^x = 30$

D. $3(0.8)^x = x$

Explain why approximation methods are needed for $2^x = x + 2$ but not for $2^x = 4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

High School — Algebra II • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • HSF-TF.A.1 • TRIGONOMETRIC FUNCTIONS

Use radian measure as arc length on the unit circle

MAIN	Algebra II main question	**HSF-TF.A.1**
BACKWARD 1	Circumference of a circle	**7.G.B.4**
BACKWARD 2	Arc length is proportional to the angle	**HSG-C.B.5**
BACKWARD 3	Degrees as fractions of a turn	**4.MD.C.5**
FORWARD 1	Measure negative angles and angles beyond one turn	**HSF-TF.A.1**
FORWARD 2	Read sine and cosine of radian angles from the unit circle	**HSF-TF.A.2**

SET 35 • HSF-TF.A.1

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

On the unit circle, a central angle subtends an arc of length 2.5. What is the radian measure of the angle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q1/5

Convert 150° to radians. Give an exact answer.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q2/5

Convert 7π/4 radians to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q3/5

A central angle of 2π/3 radians is drawn in a circle of radius 6 cm. Find the length of the arc it subtends.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q4/5

Explain why one full turn measures 2π radians.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • MAIN • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 1

Circumference of a circle • Grade 7 • Geometry • 7.G.B.4

Find the circumference of a circle of radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q1/5

Find the circumference of a circle of radius 6 cm.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q2/5

True or false?

The circumference of a circle is πr^2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q3/5

Half of the unit circle has length

A. 2π

B. 1

C. π

D. π/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q4/5

What length is one quarter of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 2

Arc length is proportional to the angle • High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5

A 60° angle subtends what fraction of a full circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q1/5

Find the arc length of a 90° angle in a circle of radius 4.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q2/5

True or false?

Doubling the radius doubles the arc length for the same angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q3/5

The arc length for angle θ (radians) in a circle of radius r is

A. θ/r

B. $r^2\theta$

C. $2\pi\theta$

D. $r\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q4/5

What fraction of the circle is an angle of π/3 radians?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

High School — Geometry (course placement in this bank) • Geometry: Circles • HSG-C.B.5 • Set 35 • BACKWARD 2 • Q5/5

SET 35 • HSF-TF.A.1

BACKWARD 3

Degrees as fractions of a turn • Grade 4 • Measurement & Data • 4.MD.C.5

How many degrees are in a full turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q1

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q1/5

What fraction of a turn is 150°?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q2

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q2/5

True or false?

A 315° angle is 3/4 of a turn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q3

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q3/5

A quarter turn is

A. 90°

B. 180°

C. 45°

D. 360°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q4

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q4/5

How many degrees are in 2/3 of a turn?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B3-Q5

Grade 4 • Measurement & Data • 4.MD.C.5 • Set 35 • BACKWARD 3 • Q5/5

SET 35 • HSF-TF.A.1

FORWARD 1

Measure negative angles and angles beyond one turn • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

Give a positive angle less than 2π that is coterminal with $-\pi/3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q1/5

Give the angle between 0 and 2π that is coterminal with 13π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q2/5

True or false?

An arc of length 10 on the unit circle goes more than once around the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q3/5

An angle of $-\pi/2$ radians is traced

A. a quarter turn clockwise

B. one full turn

C. half a turn

D. a quarter turn counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q4/5

A point travels 10 units counterclockwise around the unit circle from (1, 0). How far past one full turn is it? Give a decimal to the nearest hundredth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 35 • FORWARD 1 • Q5/5

SET 35 • HSF-TF.A.1

FORWARD 2

Read sine and cosine of radian angles from the unit circle • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

What point of the unit circle is at angle π? Give cos π and sin π.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q1/5

Find $\cos(3\pi/2)$ and $\sin(3\pi/2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q2/5

True or false?

$\cos(2\pi) = 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q3/5

$\sin(\pi/2)$ equals

A. -1

B. 0

C. $\pi/2$

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q4/5

Explain why cos θ and sin θ make sense for θ = 13π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 35 • FORWARD 2 • Q5/5

SET 36 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

Find exact trigonometric values beyond the first quadrant

MAIN	Algebra II main question	**HSF-TF.A.2**
BACKWARD 1	Special right triangle ratios	**HSG-SRT.C.6**
BACKWARD 2	Reference angles for radian measures	**HSF-TF.A.1**
BACKWARD 3	Signs of coordinates by quadrant	**6.NS.C.6.b**
FORWARD 1	Values for angles beyond 2π and solving for angles	**HSF-TF.A.2**
FORWARD 2	Verify the Pythagorean identity at exact values	**HSF-TF.C.8**

SET 36 • HSF-TF.A.2

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Use the unit circle to find sin(5π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q1/5

Use the unit circle to find cos(4π/3) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q2/5

Use the unit circle to find $\tan(7\pi/4)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q3/5

Use the unit circle to find $\sin(-\pi/2)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q4/5

Use the unit circle to find cos(11π/6) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • MAIN • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 1

Special right triangle ratios • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

In a 30°−60°−90° triangle with hypotenuse 1, give the side lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q1/5

Find sin 45° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q2/5

True or false?

$\cos 60° = \sqrt{3}/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q3/5

sin 60° equals

A. √2/2

B. √3/2

C. √3

D. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q4/5

Find tan 30° exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 2

Reference angles for radian measures • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

Find the reference angle of 5π/6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q1/5

Find the reference angle of 4π/3.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q2/5

True or false?

The reference angle of $7\pi/4$ is $3\pi/4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q3/5

Which quadrant contains 11π/6?

A. I

B. IV

C. III

D. II

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q4/5

Convert 4π/3 to degrees.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • HSF-TF.A.2

BACKWARD 3

Signs of coordinates by quadrant • Grade 6 • The Number System • 6.NS.C.6.b

In which quadrant are both coordinates negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q1/5

What are the signs of the coordinates in quadrant II?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q2/5

True or false?

A point in quadrant IV has a positive y-coordinate.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q3/5

The point $(-\sqrt{3}/2, 1/2)$ is in quadrant

A. I

B. II

C. III

D. IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q4/5

Give the signs of x and y for a point in quadrant IV.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 36 • BACKWARD 3 • Q5/5

SET 36 • HSF-TF.A.2

FORWARD 1

Values for angles beyond 2π and solving for angles • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Find $\sin(13\pi/6)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q1/5

Find $\cos(-3\pi/4)$ exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q2/5

Find every θ with $0 \le \theta < 2\pi$ and $\cos\theta = -1/2$.

$\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q3/5

True or false?

tan(π/2) is undefined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q4/5

Which angle has $\sin\theta = -\sqrt{3}/2$ and $\cos\theta = 1/2$?

A. $5\pi/3$

B. $4\pi/3$

C. $2\pi/3$

D. $7\pi/6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • HSF-TF.A.2

FORWARD 2

Verify the Pythagorean identity at exact values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

Verify $\sin^2\theta + \cos^2\theta = 1$ for $\theta = 5\pi/6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q1/5

Verify $\sin^2\theta + \cos^2\theta = 1$ for $\theta = 7\pi/4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q2/5

True or false?

$\sin^2\theta + \cos^2\theta = 1$ is true only for first-quadrant angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q3/5

If $\cos\theta = -1/2$ and θ is in quadrant III, then $\sin\theta =$

A. $\sqrt{3}/2$

B. $-\sqrt{3}/2$

C. $1/2$

D. $-1/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q4/5

Explain why the values found from the unit circle always satisfy $\sin^2\theta + \cos^2\theta = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 36 • FORWARD 2 • Q5/5

SET 37 • HSF-TF.A.2 • TRIGONOMETRIC FUNCTIONS

Explain the extension of sine and cosine to all real numbers

MAIN	Algebra II main question	**HSF-TF.A.2**
BACKWARD 1	Right-triangle definitions of sine and cosine	**HSG-SRT.C.6**
BACKWARD 2	Points on the unit circle	**HSG-GPE.A.1**
BACKWARD 3	Radian measure as directed arc length	**HSF-TF.A.1**
FORWARD 1	Explain where tangent is defined and how values repeat	**HSF-TF.A.2**
FORWARD 2	Explain symmetry and periodicity	**HSF-TF.A.4**

SET 37 • HSF-TF.A.2

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Which statement correctly explains how the unit circle defines cos t and sin t for every real number t?

A. Start at (0, 1) and travel t units clockwise; cos t is the y-coordinate and sin t is the x-coordinate of the endpoint.

B. Start at (1, 0) and travel |t| units along the unit circle, counterclockwise if $t > 0$ and clockwise if $t < 0$; cos t and sin t are the x- and y-coordinates of the endpoint.

C. Travel t units counterclockwise from (1, 0); for $t < 0$ the values are undefined.

D. Draw a right triangle with an acute angle of t radians; cos t and sin t are the ratios adjacent/hypotenuse and opposite/hypotenuse, so the definition works only for t between 0 and π/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q1/5

Which explanation shows why $\sin(t + 2\pi) = \sin t$ for every real number t?

A. For $t = \pi/2$ both sides equal 1, so they are always equal.

B. Sine distributes over addition, so $\sin(t + 2\pi) = \sin t + \sin 2\pi$, and $\sin 2\pi = 0$ because the point at 2π is $(1, 0)$; therefore $\sin(t + 2\pi) = \sin t$.

C. Adding 2π to t reflects the endpoint on the unit circle over the x-axis, and a reflection over the x-axis keeps the y-coordinate, so $\sin(t + 2\pi) = \sin t$.

D. The unit circle has circumference 2π, so adding 2π adds one full trip around it: t and $t + 2\pi$ end at the same point, which has the same y-coordinate.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q2/5

Why is $\cos t < 0$ for $\pi/2 < t < \pi$?

A. For these t the endpoint on the unit circle is in quadrant II, where x-coordinates are negative, and cos t is the x-coordinate.

B. For these t the endpoint is in quadrant II, where y-coordinates are negative, and cos t is the y-coordinate.

C. $\cos(2\pi/3) = -1/2$, so cos t is negative on this interval.

D. On the unit circle cos t decreases steadily as t increases from 0, so once t passes $\pi/2$, where $\cos t = 0$, the value of cos t is negative for every larger t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q3/5

Why is $-1 \le \sin t \le 1$ for every real number t?

A. $\sin(\pi/2) = 1$ and $\sin(3\pi/2) = -1$, so every value lies between them.

B. In a right triangle the hypotenuse is longer than either leg, so opposite/hypotenuse is less than 1, which means $\sin t < 1$ for every real number t.

C. sin t is the y-coordinate of a point on the unit circle $x^2 + y^2 = 1$, so $y^2 \le 1$, which means $-1 \le y \le 1$.

D. sin t is an arc length on the unit circle, and those arcs are at most 1.

How does the unit-circle definition give $\sin(-\pi/6)$?

A. Travel $\pi/6$ clockwise from $(1, 0)$ to $(1/2, -\sqrt{3}/2)$, so $\sin(-\pi/6) = -\sqrt{3}/2$.

B. Negative inputs are not on the unit circle, so $\sin(-\pi/6)$ is undefined.

C. Travel $\pi/6$ clockwise from $(1, 0)$ to the point $(\sqrt{3}/2, -1/2)$, so $\sin(-\pi/6) = -1/2$.

D. Travel $\pi/6$ counterclockwise from $(1, 0)$ to the point $(\sqrt{3}/2, 1/2)$, so $\sin(-\pi/6) = 1/2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • MAIN • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 1

Right-triangle definitions of sine and cosine • High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

Define sin A in a right triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q1/5

In a right triangle with hypotenuse 1, what are the legs next to and opposite an acute angle θ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q2/5

True or false?

The right-triangle definition works for $\theta = 120°$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q3/5

In a right triangle with hypotenuse 1, the leg opposite θ is

A. $\sin \theta$

B. $\tan \theta$

C. 1

D. $\cos \theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-B1-Q4**

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q4/5

Why does the right-triangle definition fail for negative angles?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

High School — Geometry (course placement in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 37 • BACKWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 2

Points on the unit circle • High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Write the equation of the circle with center (0, 0) and radius 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q1/5

Is $(-1/2, \sqrt{3}/2)$ on the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q2/5

True or false?

Every point on the unit circle has $-1 \leq x \leq 1$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations
HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q3/5

Which point is on the unit circle?

A. (0.9, 0.9)

B. (0.5, 0.5)

C. (1, 1)

D. (0.6, −0.8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q4/5

A point on the unit circle has x = 0. Give both possible points.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

High School — Geometry (course placement in this bank) • Geometry: Expressing Geometric Properties with Equations HSG-GPE.A.1 • Set 37 • BACKWARD 2 • Q5/5

SET 37 • HSF-TF.A.2

BACKWARD 3

Radian measure as directed arc length • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1

What arc length on the unit circle does t = π/6 cut off?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q1/5

Which direction does $t = -\pi/6$ travel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q2/5

True or false?

$t = \pi$ returns to the starting point (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q3/5

An arc of length π from (1, 0) ends at

A. (0, 1)

B. (0, −1)

C. (1, 0)

D. (−1, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q4/5

Give a negative t that ends at $(0, -1)$.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.1 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • HSF-TF.A.2

FORWARD 1

Explain where tangent is defined and how values repeat • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Why is tan t undefined at $t = \pi/2 + k\pi$ (k an integer)?

A. At these t the endpoint on the unit circle is (0, 1) or (0, −1), so cos t = 0 and tan t = sin t/cos t would divide by 0.

B. tan t is undefined whenever sin t = 1.

C. At these t the endpoint on the unit circle is (1, 0) or (−1, 0), so sin t = 0 and tan t = sin t/cos t would have to divide by 0.

D. tan t = cos t/sin t, and sin t = 0 at these t.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q1/5

Give the domain and range of $y = \sin t$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q2/5

True or false?

cos t and sin t can both equal 0 for the same t.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q3/5

The unit-circle definition agrees with the triangle definition for

A. $t < 0$ only

B. $\pi/2 < t < \pi$ only

C. all t

D. $0 < t < \pi/2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q4/5

Explain why $\cos t = \cos(-t)$ using the unit circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 37 • FORWARD 1 • Q5/5

SET 37 • HSF-TF.A.2

FORWARD 2

Explain symmetry and periodicity • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4

Which unit-circle argument shows that $\sin(-t) = -\sin t$?

A. $\sin(-\pi/2) = -1 = -\sin(\pi/2)$, so the rule holds for every t.

B. The endpoints for t and $-t$ are reflections of each other over the x-axis, so their y-coordinates (the sines) are opposites.

C. The endpoints for t and $-t$ are the same point, so their sines are opposites.

D. The endpoints for t and $-t$ are always reflections of each other over the y-axis, so their y-coordinates (the sines) are opposites.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S37-F2-Q1**

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q1/5

Is sine an even function, an odd function, or neither?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q2/5

True or false?

$\tan(t + \pi) = \tan t$ for every t where it is defined.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q3/5

The period of $y = \tan t$ is

A. π

B. 4π

C. $\pi/2$

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q4/5

Explain why cos is an even function.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.A.4 • Set 37 • FORWARD 2 • Q5/5

SET 38 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

Write sinusoidal models with given amplitude, period, and midline

MAIN	Algebra II main question	**HSF-TF.B.5**
BACKWARD 1	Amplitude and midline from the maximum and minimum	**7.NS.A.3**
BACKWARD 2	Period and frequency	**6.RP.A.2**
BACKWARD 3	Vertical stretches, shifts, and reflections	**HSF-BF.B.3**
FORWARD 1	Write models from data or with a phase shift	**HSF-TF.B.5**
FORWARD 2	Solve for times in sinusoidal models	**HSF-TF.B.7**

SET 38 • HSF-TF.B.5

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Write a cosine function with amplitude 3, period 2π, midline $y = 1$, and a maximum at $x = 0$.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q1/5

A Ferris wheel has diameter 40 m, its center is 25 m above the ground, and it turns once every 4 minutes. A rider starts at the bottom. Write the rider's height h after t minutes.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q2/5

Water depth at a harbor varies between 1 ft (low tide) and 9 ft (high tide) with a period of 12 hours. High tide is at t = 0. Write the depth d after t hours.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q3/5

A town's average temperature is lowest, 10°C, in month t = 0 and highest, 30°C, at t = 6, with a 12-month period. Write a model T(t).

T = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q4/5

Write a sine function with amplitude 2, period π, and midline $y = -1$ that is at the midline and increasing at $x = 0$.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • MAIN • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 1

Amplitude and midline from the maximum and minimum • Grade 7 • The Number System • 7.NS.A.3

Find (9 + 1)/2 and (9 − 1)/2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q1/5

A quantity varies between 10 and 30. Find the midline value and amplitude.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q2/5

True or false?

The amplitude is the maximum minus the minimum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q3/5

For max 45 and min 5, the midline is

A. y = 20

B. y = 50

C. y = 25

D. y = 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q4/5

A Ferris wheel has diameter 40 m with center 25 m high. Give its maximum and minimum heights.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 38 • BACKWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 2

Period and frequency • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2

A wheel turns once every 4 minutes. How many turns per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q1/5

A spring oscillates 3 times per second. What is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q2/5

True or false?

Frequency is equal to the period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q3/5

A period of 12 hours means a frequency of

A. 12 cycles per hour

B. 2 cycles per hour

C. 1/2 cycle per hour

D. 1/12 cycle per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q4/5

$y = \sin(Bx)$ has period $2\pi/B$. Find B for period 4.

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 38 • BACKWARD 2 • Q5/5

SET 38 • HSF-TF.B.5

BACKWARD 3

Vertical stretches, shifts, and reflections • High School — Algebra I • Functions: Building Functions • HSF-BF.B.3

Describe how $y = 3\cos x + 1$ is obtained from $y = \cos x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q1

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q1/5

Describe $y = -\cos x$ compared with $y = \cos x$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q2

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q2/5

True or false?

$y = \cos x + 5$ has a maximum of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q3

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q3/5

$y = -20\cos x$ starts (at $x = 0$) at

A. its minimum

B. its maximum

C. the midline going up

D. the midline going down

Write y = sin x shifted down 1 and stretched vertically by 2.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

High School — Algebra I • Functions: Building Functions • HSF-BF.B.3 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • HSF-TF.B.5

FORWARD 1

Write models from data or with a phase shift • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

A rider's height is 3 m at t = 0 (bottom), 27 m at the top after 15 s, and the motion repeats every 30 s. Write h(t).

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q1/5

Sunrise times vary between 5:30 and 7:30 (as 5.5 and 7.5 hours) with period 365 days, latest at day 0. Write a model.

S = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q2/5

True or false?

$y = 4\sin(x - \pi/2)$ is the same as $y = 4\cos x$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q3/5

A sine model with period 10, amplitude 3, midline 7, and a maximum at t = 2.5 is

A. $7 + 3\sin(10t)$

B. $7 + 3\sin(\pi t/5)$

C. $3 + 7\sin(\pi t/5)$

D. $7 + 3\cos(\pi t/5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q4/5

A model has a maximum of 18 at t = 3 and the next minimum of 6 at t = 7. Write a cosine model.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 38 • FORWARD 1 • Q5/5

SET 38 • HSF-TF.B.5

FORWARD 2

Solve for times in sinusoidal models • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

For $h = 25 - 20\cos(\pi t/2)$, find the first time $t > 0$ when the rider is at 35 m. A calculator may be used.

$t \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q1/5

For $d = 4\cos(\pi t/6) + 5$, find all times in $0 \leq t < 12$ when the depth is 5 ft.

$t =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q2/5

True or false?

A sinusoidal model usually reaches a given value between its min and max twice per period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q3/5

For $T = 20 - 10\cos(\pi t/6)$, $T = 25$ first when

A. $t = 2$

B. $t = 4$

C. $t = 3$

D. $t = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q4/5

Explain why there are two solutions per period when the value is strictly between the minimum and maximum.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 38 • FORWARD 2 • Q5/5

SET 39 • HSF-TF.B.5 • TRIGONOMETRIC FUNCTIONS

Interpret the parameters of sinusoidal models

MAIN	Algebra II main question	**HSF-TF.B.5**
BACKWARD 1	Build the model from amplitude, period, and midline	**HSF-TF.B.5**
BACKWARD 2	Evaluate a sinusoidal model	**HSF-TF.A.2**
BACKWARD 3	Convert rates between units	**6.RP.A.3.d**
FORWARD 1	Predict how changing a parameter changes the situation	**HSF-TF.B.5**
FORWARD 2	Interpret periodic features from graphs and tables	**HSF-IF.B.4**

SET 39 • HSF-TF.B.5

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ meters after t seconds. Give the maximum height, the minimum height, and the time for one revolution.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q1/5

Hours of daylight are $D(t) = 12 + 3\sin(2\pi(t - 80)/365)$, t in days. Interpret 12, 3, and 365.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q2/5

A model of blood pressure is $P(t) = 100 + 20\sin(2.5\pi t)$, t in seconds. Find the period and the number of heartbeats per minute.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q3/5

A spring's position is $y = -4\cos(3\pi t)$ cm, t in seconds. Give the amplitude, period, and frequency.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q4/5

Two sounds are $y = \sin(880\pi t)$ and $y = \sin(440\pi t)$, t in seconds. Give the frequency of each in cycles per second and say which is higher.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • MAIN • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 1

Build the model from amplitude, period, and midline • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

What is the amplitude of $y = 12 + 10\sin(\pi t/15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q1/5

What is the midline of $y = 12 + 10\sin(\pi t/15)$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q2/5

True or false?

The period of $y = \sin(Bt)$ is $B/(2\pi)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q3/5

The period of $y = \sin(\pi t/15)$ is

A. 30

B. 15

C. $\pi/15$

D. 2π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q4/5

Write a model with amplitude 4, period 2/3, and midline 0.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 2

Evaluate a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Evaluate $12 + 10\sin(\pi/2)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q1/5

Evaluate 12 + 10sin(3π/2).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q2/5

True or false?

$-4\cos(0) = 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q3/5

100 + 20sin(π/2) equals

A. 100

B. 20

C. 120

D. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q4/5

Evaluate 12 + 3sin(0).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 39 • BACKWARD 2 • Q5/5

SET 39 • HSF-TF.B.5

BACKWARD 3

Convert rates between units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

A cycle lasts 0.8 s. How many cycles are in 60 s?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q1/5

1.5 cycles per second is how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q2/5

True or false?

A period of 30 s means 30 revolutions per minute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q3/5

440 cycles per second is how many cycles per millisecond?

A. 4.4

B. 44

C. 440

D. 0.44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q4/5

A tide has period 12 hours. How many cycles in 3 days?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 39 • BACKWARD 3 • Q5/5

SET 39 • HSF-TF.B.5

FORWARD 1

Predict how changing a parameter changes the situation • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

A Ferris wheel rider's height is $h(t) = 12 + 10\sin(\pi t/15)$ m after t seconds. The wheel is replaced by one of the same size that turns twice as fast. Rewrite h(t).

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q1/5

Daylight in one city is $D(t) = 12 + 3\sin(2\pi(t - 80)/365)$ hours. A city farther from the equator has daylight between 8 and 16 hours with the same period and timing. Which parameter changes, and to what?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q2/5

True or false?

Changing the midline changes the period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q3/5

Multiplying B by 3 makes the period

A. 3 units longer

B. unchanged

C. three times as long

D. one third as long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q4/5

A heart rate rises from 75 to 100 beats per minute. What is the new period, in seconds?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 39 • FORWARD 1 • Q5/5

SET 39 • HSF-TF.B.5

FORWARD 2

Interpret periodic features from graphs and tables • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

A table of a tide's depth shows maxima at t = 2 and t = 14 hours. What is the period?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q1/5

A graph of a temperature model has a high of 30 and a low of 10. What are the midline and amplitude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q2/5

True or false?

Consecutive minima of a periodic function are one period apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q3/5

On which interval does $h(t) = 12 + 10\sin(\pi t/15)$ increase?

A. $0 < t < 15$

B. $0 < t < 7.5$

C. $15 < t < 30$

D. $7.5 < t < 22.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q4/5

Explain how to read the frequency of a periodic graph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 39 • FORWARD 2 • Q5/5

SET 40 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

Prove the Pythagorean identity

MAIN	Algebra II main question	**HSF-TF.C.8**
BACKWARD 1	The Pythagorean Theorem	**8.G.B.7**
BACKWARD 2	Coordinates of the point at angle θ	**HSF-TF.A.2**
BACKWARD 3	Expand and simplify algebraic squares	**HSA-SSE.A.2**
FORWARD 1	Prove identities derived from the Pythagorean identity	**HSF-TF.C.8**
FORWARD 2	Use the angle-addition formulas	**HSF-TF.C.9**

SET 40 • HSF-TF.C.8

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

Which argument proves $\sin^2\theta + \cos^2\theta = 1$ for every angle θ, using the unit circle?

A. On the unit circle $\sin\theta + \cos\theta = 1$ for every angle θ, because the coordinates of a point add to the radius; squaring both sides gives $\sin^2\theta + \cos^2\theta = 1$.

B. The point at angle θ is $(\sin\theta, \cos\theta)$, so $\sin\theta + \cos\theta = 1$.

C. For $\theta = 30°$: $(1/2)^2 + (\sqrt{3}/2)^2 = 1$, so the identity holds.

D. The point at angle θ is $(\cos\theta, \sin\theta)$, and every point on the unit circle satisfies $x^2 + y^2 = 1$; substituting gives $\cos^2\theta + \sin^2\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q1/5

A right triangle has legs a and b, hypotenuse c, and acute angle θ opposite a. Which argument proves $\sin^2\theta + \cos^2\theta = 1$?

A. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a + b)^2/c^2$, and $(a + b)^2 = c^2$ by the Pythagorean Theorem, so the sum is 1.

B. $\sin\theta = b/c$ and $\cos\theta = a/c$, so $\sin\theta + \cos\theta = (a + b)/c = 1$.

C. $\sin\theta = a/c$ and $\cos\theta = b/c$, so $\sin^2\theta + \cos^2\theta = (a^2 + b^2)/c^2 = c^2/c^2 = 1$ by the Pythagorean Theorem.

D. In a 3-4-5 triangle, $(3/5)^2 + (4/5)^2 = 1$, so the identity holds.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q2/5

Which argument proves $1 + \tan^2\theta = 1/\cos^2\theta$ wherever $\cos\theta \neq 0$?

A. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\sin^2\theta$ to get $1 + \cos^2\theta/\sin^2\theta = 1/\sin^2\theta$; since $\cos^2\theta/\sin^2\theta = \tan^2\theta$, this gives $1 + \tan^2\theta = 1/\cos^2\theta$.

B. For $\theta = 0$: $1 + 0 = 1/1$, so the identity holds.

C. Square $\tan\theta = \sin\theta/\cos\theta$ to get $\tan^2\theta = \sin^2\theta/\cos^2\theta = 1/\cos^2\theta$.

D. Divide $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$, which is allowed because $\cos\theta \neq 0$: $\sin^2\theta/\cos^2\theta + 1 = 1/\cos^2\theta$, and $\sin^2\theta/\cos^2\theta = \tan^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q3/5

θ is in quadrant II, and its point on the unit circle is (x, y) with $x < 0 < y$. Which argument proves $\sin^2\theta + \cos^2\theta = 1$ with a reference triangle?

A. The reference triangle has legs x and y, so $x + y = 1$, which means $\cos\theta + \sin\theta = 1$.

B. The reference triangle has legs $|x|$ and $|y|$ and hypotenuse 1, so $|x|^2 + |y|^2 = 1$. Squaring removes the signs, so $x^2 + y^2 = \cos^2\theta + \sin^2\theta = 1$.

C. The legs of the reference triangle are x and y, and x is negative in quadrant II, so $x^2 + y^2 = -1$ there, which means $\cos^2\theta + \sin^2\theta = -1$ instead of 1.

D. The identity was proved for acute angles, so it is true in quadrant II with no further work.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-M-Q4**

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q4/5

Which argument proves $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$?

A. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + 2\sin\theta\cos\theta + \cos^2\theta$, and $\sin^2\theta + \cos^2\theta = 1$.

B. $(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta = 1$, so the term $2\sin\theta\cos\theta$ is extra.

C. For $\theta = 0$: $(0 + 1)^2 = 1 + 0$, so the identity holds.

D. $(\sin\theta + \cos\theta)^2 = 1^2 = 1$ because $\sin\theta + \cos\theta = 1$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • MAIN • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 1

The Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs 3 and 4. Find the hypotenuse.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q1/5

A right triangle has hypotenuse 1 and a leg 0.6. Find the other leg.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q2/5

True or false?

In a right triangle, $a^2 + b^2 = c^2$ where c is the hypotenuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q3/5

Legs 5 and 12 give hypotenuse

A. 60

B. 7

C. 17

D. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q4/5

Legs a and b and hypotenuse c: write c^2 in terms of a and b.

$c^2 =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 2

Coordinates of the point at angle θ • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

Give the coordinates of the point on the unit circle at angle θ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q1/5

Give $\cos(2\pi/3)$ and $\sin(2\pi/3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q2/5

True or false?

sin θ is the x-coordinate of the point at angle θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q3/5

The point at angle π/4 is

A. (1/2, √3/2)

B. (1, 1)

C. (√3/2, 1/2)

D. (√2/2, √2/2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q4/5

What is the radius of the unit circle?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • HSF-TF.C.8

BACKWARD 3

Expand and simplify algebraic squares • High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2

Expand $(a + b)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q1

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q1/5

Simplify $(a/c)^2 + (b/c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q2

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q2/5

True or false?

$\sin^2\theta$ means $\sin(\theta^2)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q3/5

Dividing $x^2 + y^2 = 1$ by x^2 gives

A. $1 + y^2/x^2 = 1/x^2$

B. $1 + y^2 = 1/x$

C. $y^2 = 1$

D. $x^2 + y^2 = 1/x^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q4

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q4/5

Expand $(s + c)^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q5

High School — Algebra I • Algebra: Seeing Structure in Expressions • HSA-SSE.A.2 • Set 40 • BACKWARD 3 • Q5/5

SET 40 • HSF-TF.C.8

FORWARD 1

Prove identities derived from the Pythagorean identity • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

Which argument proves $\sin^4\theta - \cos^4\theta = \sin^2\theta - \cos^2\theta$?

A. Divide both sides by $\sin^2\theta - \cos^2\theta$ to get $\sin^2\theta + \cos^2\theta = 1$, which is the Pythagorean identity, so the identity is true.

B. $\sin^4\theta - \cos^4\theta = (\sin\theta - \cos\theta)^4$, which equals $\sin^2\theta - \cos^2\theta$.

C. For $\theta = \pi/4$ both sides are 0, so the identity holds.

D. $\sin^4\theta - \cos^4\theta = (\sin^2\theta - \cos^2\theta)(\sin^2\theta + \cos^2\theta) = (\sin^2\theta - \cos^2\theta)(1) = \sin^2\theta - \cos^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q1/5

Which argument proves $(1 - \cos\theta)(1 + \cos\theta) = \sin^2\theta$?

A. For $\theta = \pi/2$: $(1 - 0)(1 + 0) = 1 = \sin^2(\pi/2)$, so the identity holds.

B. $(1 - \cos\theta)(1 + \cos\theta) = 1 - 2\cos\theta + \cos^2\theta$, and by the Pythagorean identity this equals $\sin^2\theta$.

C. $(1 - \cos\theta)(1 + \cos\theta) = 1 - \cos^2\theta$, and $1 - \cos^2\theta = \sin^2\theta$ by the Pythagorean identity.

D. $(1 - \cos\theta)(1 + \cos\theta) = 1 + \cos^2\theta$, which equals $\sin^2\theta$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q2/5

True or false?

$\sin\theta + \cos\theta = 1$ for every θ.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q3/5

Which equals $1 - \sin^2\theta$?

A. $\tan^2\theta$

B. $1 - \cos^2\theta$

C. $\cos\theta$

D. $\cos^2\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q4/5

Which argument proves $\cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$?

A. Replace $\sin^2\theta$ with $1 - \cos^2\theta$:
$\cos^2\theta - (1 - \cos^2\theta) = \cos^2\theta - 1 + \cos^2\theta = 2\cos^2\theta - 1$.

B. Add $\sin^2\theta$ to both sides to get $\cos^2\theta = 2\cos^2\theta - 1 + \sin^2\theta$, which is always true.

C. For $\theta = 0$: $1 - 0 = 2 - 1$, so the identity holds.

D. Replace $\sin^2\theta$ with $1 - \cos^2\theta$: $\cos^2\theta - 1 - \cos^2\theta = -1$, so the identity is false for every θ.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 40 • FORWARD 1 • Q5/5

SET 40 • HSF-TF.C.8

FORWARD 2

Use the angle-addition formulas • High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9

Use sin(a + b) = sin a cos b + cos a sin b to find sin(75°) exactly.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q1/5

Use $\cos(a + b) = \cos a \cos b - \sin a \sin b$ with $a = b = \theta$ to write $\cos 2\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q2/5

True or false?

$\cos 2\theta = 1 - 2\sin^2\theta$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q3/5

$\sin 2\theta$ equals

A. $\sin^2\theta - \cos^2\theta$

B. $2\sin\theta$

C. $2\sin\theta\cos\theta$

D. $\sin\theta + \cos\theta$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S40-F2-Q4**

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q4/5

Use $(\sin\theta + \cos\theta)^2 = 1 + 2\sin\theta\cos\theta$ to write $(\sin\theta + \cos\theta)^2$ in terms of $\sin 2\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

High School — Precalculus (course placement in this bank) • Functions: Trigonometric Functions • HSF-TF.C.9 • Set 40 • FORWARD 2 • Q5/5

SET 41 • HSF-TF.C.8 • TRIGONOMETRIC FUNCTIONS

Use the Pythagorean identity to find trigonometric values

MAIN	Algebra II main question	**HSF-TF.C.8**
BACKWARD 1	Signs of sine, cosine, and tangent by quadrant	**HSF-TF.A.2**
BACKWARD 2	Square roots of fractions	**8.EE.A.2**
BACKWARD 3	Solve $x^2 = c$	**HSA-REI.B.4.b**
FORWARD 1	Find values when the quadrant is not given, and combine values	**HSF-TF.C.8**
FORWARD 2	Find the angles with inverse trigonometric functions	**HSF-TF.B.7**

SET 41 • HSF-TF.C.8

MAIN

Algebra II main question • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

$\sin \theta = 3/5$ and θ is in quadrant II. Find $\cos \theta$ and $\tan \theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q1/5

$\cos\theta = -5/13$ and θ is in quadrant III. Find $\sin\theta$ and $\tan\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q2/5

$\tan\theta = -2$ and θ is in quadrant IV. Find $\sin\theta$ and $\cos\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q3/5

$\sin\theta = -1/3$ and θ is in quadrant IV. Find $\cos\theta$ and $\tan\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q4/5

$\cos\theta = 0.6$ and $\sin\theta < 0$. Find $\sin\theta$ and $\tan\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • MAIN • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 1

Signs of sine, cosine, and tangent by quadrant • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2

In quadrant II, which of sin, cos, tan are positive?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q1/5

In quadrant III, what is the sign of tan θ?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q2/5

True or false?

In quadrant IV, $\cos\theta < 0$ and $\sin\theta > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q3/5

If $\sin\theta < 0$ and $\cos\theta > 0$, θ is in quadrant

A. I

B. II

C. IV

D. III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q4/5

If $\tan\theta < 0$ and $\sin\theta > 0$, which quadrant is θ in?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.A.2 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 2

Square roots of fractions • Grade 8 • Expressions & Equations • 8.EE.A.2

Find √(16/25).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q1/5

Find √(8/9) in simplest radical form.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q2/5

True or false?

$\sqrt{(144/169)} = 12/169$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q3/5

√(1/5) equals

A. 5

B. √5

C. 1/25

D. √5/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q4/5

Find $\sqrt{0.64}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • HSF-TF.C.8

BACKWARD 3

Solve $x^2 = c$ • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b

Solve $x^2 = 16/25$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q1/5

Solve $x^2 = 1 - 9/25$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q2/5

True or false?

$x^2 = 0.36$ has only the solution 0.6.

A. True

B. False

$x^2 + (3/5)^2 = 1$ gives x =

A. ±2/5

B. 2/5

C. 4/5

D. ±4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q4/5

Solve $5x^2 = 1$.

$x =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.4.b • Set 41 • BACKWARD 3 • Q5/5

SET 41 • HSF-TF.C.8

FORWARD 1

Find values when the quadrant is not given, and combine values • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8

$\sin\theta = 3/5$. List both possible values of $\cos\theta$ and the quadrants they go with.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q1/5

$\cos\theta = -5/13$ and θ is in quadrant III. Find $\sin\theta \cdot \cos\theta$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q2/5

True or false?

Knowing $\tan \theta = 1$ determines $\sin \theta$ uniquely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q3/5

If $\sin\theta = 0.8$ and $\tan\theta < 0$, then $\cos\theta =$

A. -0.8

B. 0.2

C. 0.6

D. -0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q4/5

Use $1 + \tan^2\theta = 1/\cos^2\theta$ to find $\cos\theta$ if $\tan\theta = 3$ and θ is in quadrant III.

$\cos\theta =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.C.8 • Set 41 • FORWARD 1 • Q5/5

SET 41 • HSF-TF.C.8

FORWARD 2

Find the angles with inverse trigonometric functions • High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7

$\sin\theta = 3/5$ with $0 \leq \theta < 2\pi$. Find both values of θ to the nearest thousandth. A calculator may be used.

$\theta \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q1/5

$\cos\theta = -5/13$ and θ is in quadrant III. Find θ to the nearest thousandth ($0 \leq \theta < 2\pi$). A calculator may be used.

$\theta \approx$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q2/5

True or false?

arcsin gives only angles between $-\pi/2$ and $\pi/2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q3/5

The quadrant II angle with $\sin\theta = 3/5$ is

A. $\pi + \arcsin(3/5)$

B. $\pi - \arcsin(3/5)$

C. $\arcsin(3/5)$

D. $-\arcsin(3/5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q4/5

Explain why the identity finds cos θ but the inverse function is needed to find θ itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

High School — Algebra II, Advanced Extension (+) • Functions: Trigonometric Functions • HSF-TF.B.7 • Set 41 • FORWARD 2 • Q5/5

SET 42 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret key features of polynomial and rational models

MAIN	Algebra II main question	**HSF-IF.B.4**
BACKWARD 1	Interpret key features of linear and quadratic models	**HSF-IF.B.4**
BACKWARD 2	Describe increasing and decreasing behavior	**8.F.B.5**
BACKWARD 3	Zeros and sign from factors	**HSA-APR.B.3**
FORWARD 1	Sketch a model from a verbal description of key features	**HSF-IF.B.4**
FORWARD 2	Compare key features across representations	**HSF-IF.C.9**

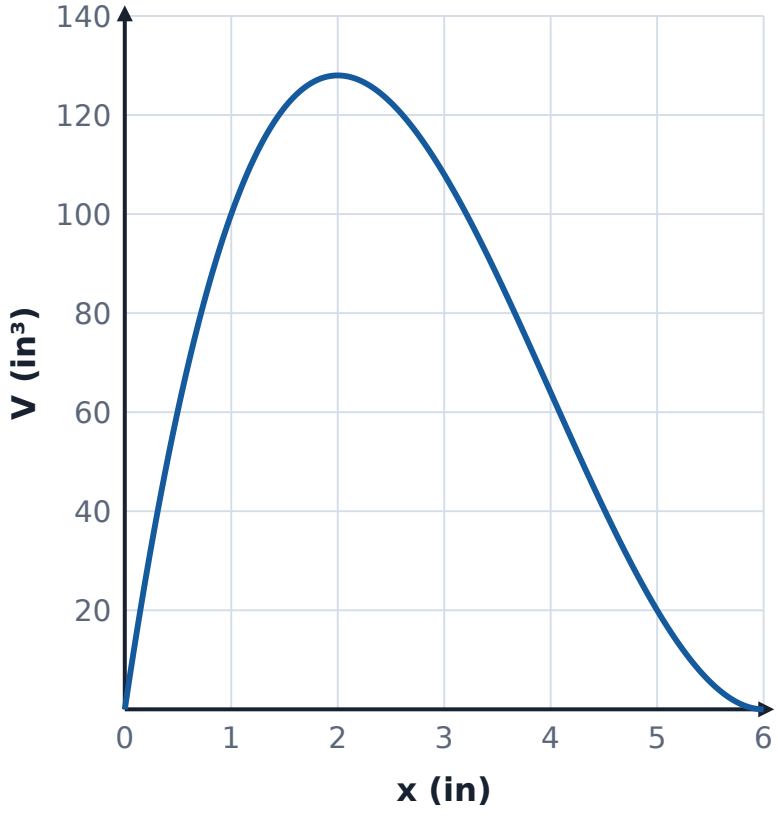
SET 42 • HSF-IF.B.4

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

An open box is made from a 12-in square sheet by cutting squares of side x from the corners. Its volume is $V(x) = x(12 - 2x)^2$. Use the graph to give the x that maximizes V, the maximum volume, and the interval where V increases.

Answer: _____

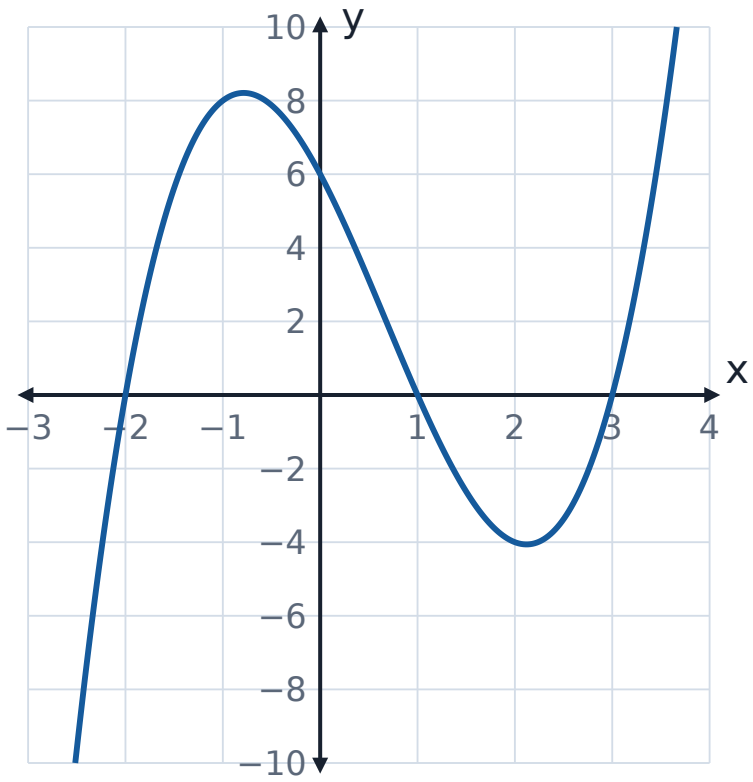


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q1/5

The graph of $f(x) = (x + 2)(x - 1)(x - 3)$ is shown. Give the intervals where $f(x) > 0$ and the intervals where $f(x) < 0$.



Answer: _____

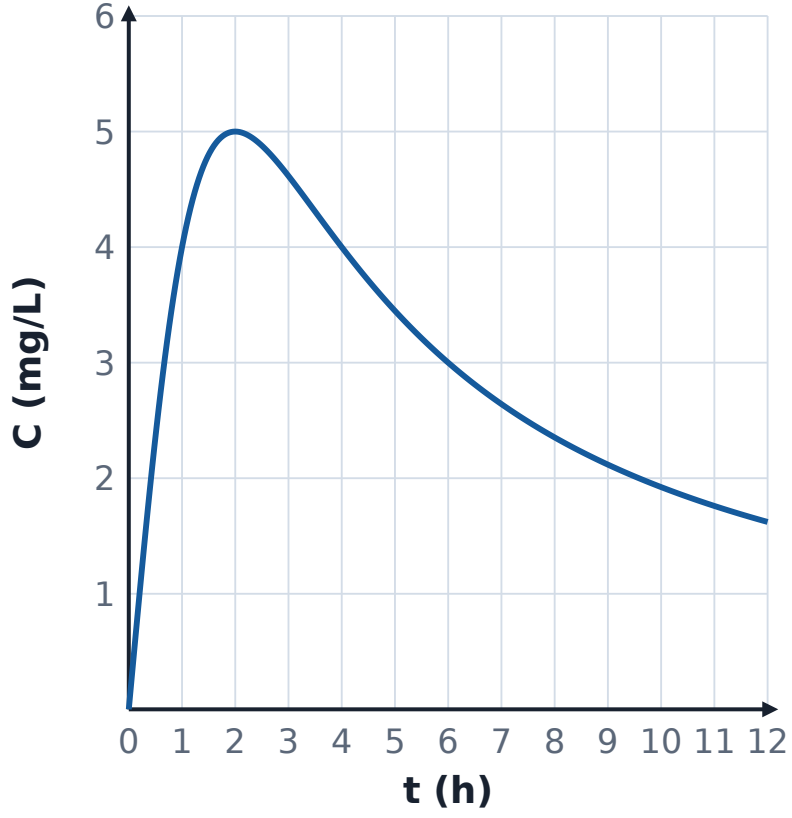
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q2/5

A drug concentration is $C(t) = 20t/(t^2 + 4)$ mg/L after t hours. Use the graph to give the maximum concentration, when it occurs, and the end behavior. Interpret the end behavior.

Answer: _____

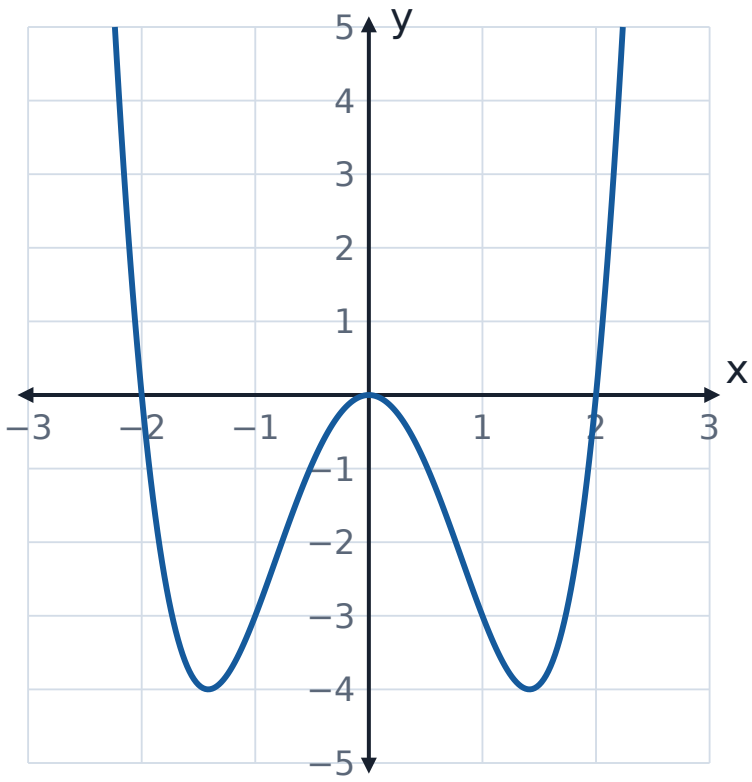


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q3/5

The graph of $f(x) = x^4 - 4x^2$ is shown. Give its x-intercepts and its relative maximum and minimum points.



Answer: _____

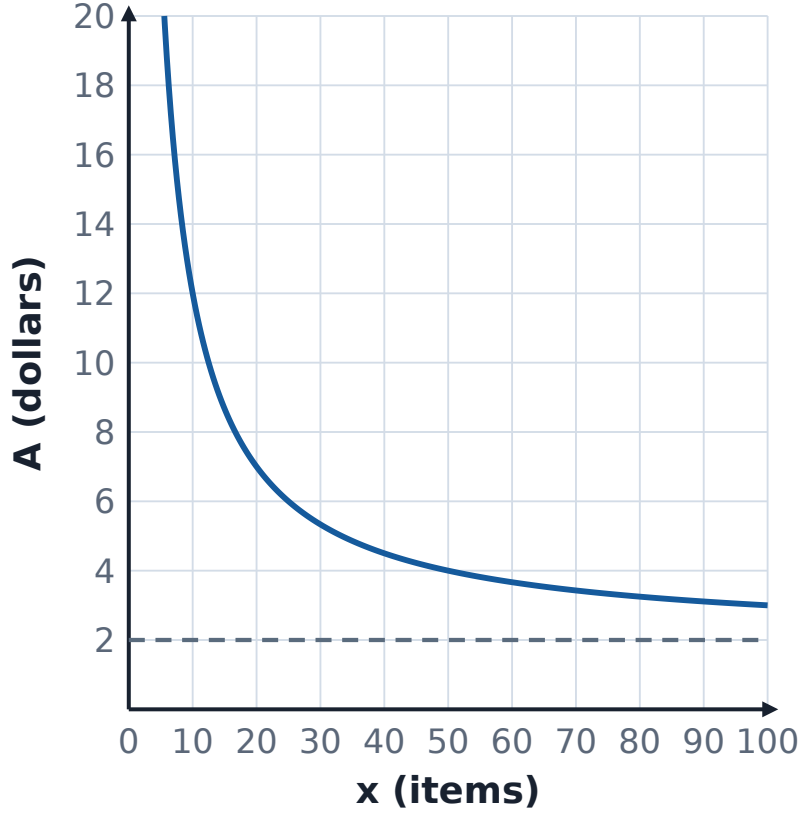
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q4/5

The average cost of making x items is A(x) = (100 + 2x)/x dollars. Use the graph to describe the end behavior and interpret it.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • MAIN • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 1

Interpret key features of linear and quadratic models • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

$h(t) = -16t^2 + 32t + 4$. When is the maximum height, and what is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q1/5

What does the y-intercept of C = 25 + 3x mean if C is the cost of x rides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q2/5

True or false?

A parabola that opens down has a minimum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q3/5

$h(t) = -16t^2 + 32t + 4$ increases for

A. all t

B. $0 < t < 1$

C. $t > 1$

D. $1 < t < 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q4/5

Where is $y = (x - 2)(x - 6)$ negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 2

Describe increasing and decreasing behavior • Grade 8 • Functions • 8.F.B.5

A graph rises from x = 0 to x = 2 and then falls. Describe it in words.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q1/5

Sketch-free: describe the graph of the concentration that rises then levels toward 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q2/5

True or false?

A highest point between rising and falling parts is a relative maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q3/5

A function that falls on $1 < x < 3$ is

A. constant there

B. decreasing there

C. undefined there

D. increasing there

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q4/5

What is a relative minimum?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 8 • Functions • 8.F.B.5 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • HSF-IF.B.4

BACKWARD 3

Zeros and sign from factors • High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3

List the zeros of $(x + 2)(x - 1)(x - 3)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q1/5

List the zeros of $x^4 - 4x^2 = x^2(x - 2)(x + 2)$ with multiplicities.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q2/5

True or false?

$V(x) = x(12 - 2x)^2$ has zeros 0 and 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q3

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q3/5

$20t/(t^2 + 4) = 0$ when

A. never

B. $t = -2$

C. $t = 2$

D. $t = 0$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q4

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q4/5

Is $(x + 2)(x - 1)(x - 3)$ positive or negative at $x = 2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q5

High School — Algebra II • Algebra: Arithmetic with Polynomials & Rational Expressions • HSA-APR.B.3 • Set 42 • BACKWARD 3 • Q5/5

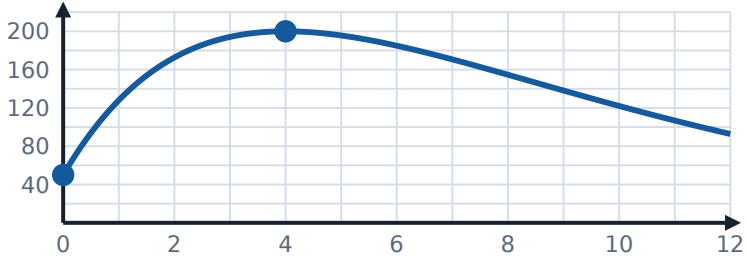
SET 42 • HSF-IF.B.4

FORWARD 1

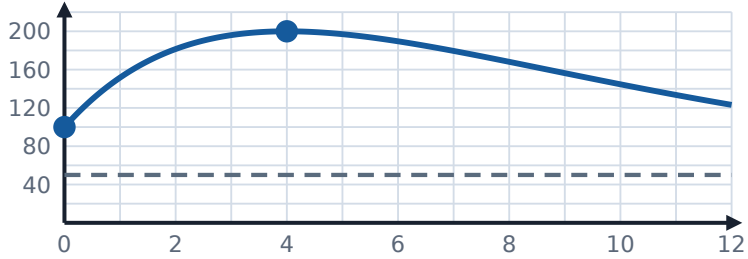
Sketch a model from a verbal description of key features • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

Which graph could show a population that is 50 at t = 0, rises to a maximum of 200 at t = 4, and then decreases toward a level of 100 as t grows?

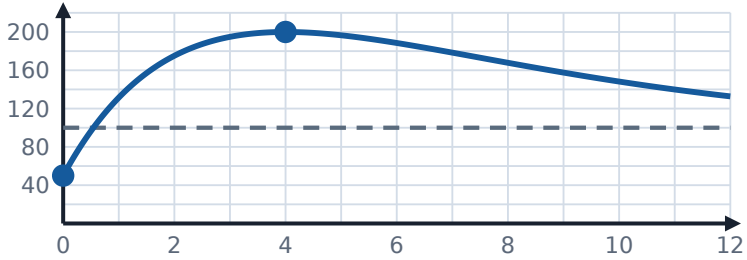
A



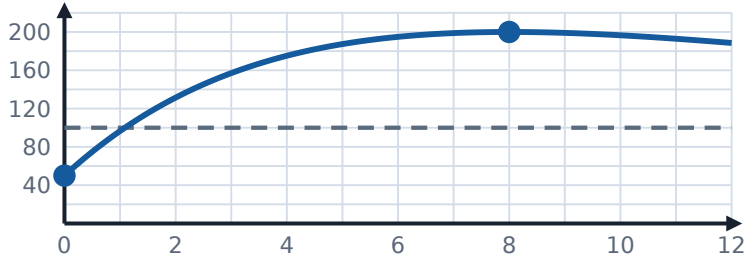
B



C



D



A. Graph A

B. Graph B

C. Graph C

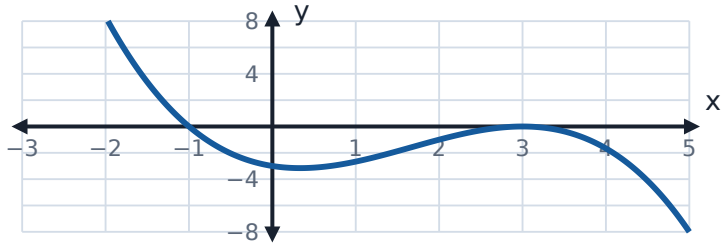
D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q1**

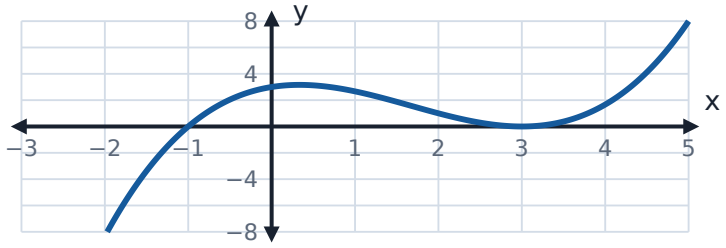
High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q1/5

Which graph could show a cubic with zeros at −1 and 3 only, a negative y-intercept, falling to the right, and touching the x-axis at 3?

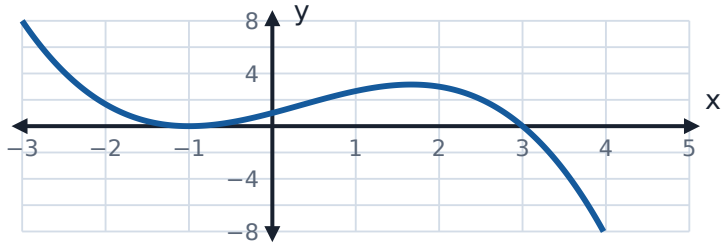
A



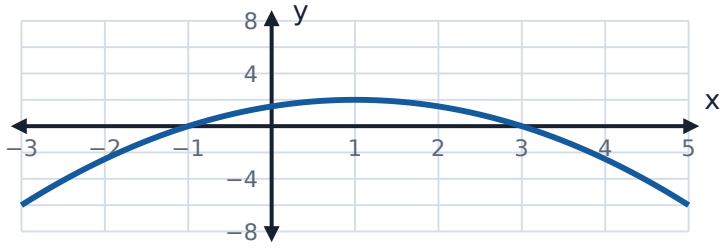
B



C



D



A. Graph A

B. Graph B

C. Graph C

D. Graph D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F1-Q2**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q2/5

True or false?

A graph can have a relative maximum that is not the absolute maximum.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q3/5

A model rises to a peak and approaches 0 as t grows. Which formula fits?

A. $20t/(t^2 + 4)$

B. $20t$

C. t^2

D. $20 - t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q4/5

Write a polynomial with zeros 0 and 6 and a relative maximum in between, whose values are positive between the zeros.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 42 • FORWARD 1 • Q5/5

SET 42 • HSF-IF.B.4

FORWARD 2

Compare key features across representations • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9

$f(x) = x(12 - 2x)^2$ and g is a quadratic with vertex (3, 120). Which has the greater maximum on $0 < x < 6$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q1/5

$C(t) = 20t/(t^2 + 4)$ and a table of $D(t)$ gives $D(1) = 3$, $D(2) = 6$, $D(3) = 4$. Which has the greater peak among these values, and when?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q2/5

True or false?

Two functions with the same zeros have the same relative extrema.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q3/5

Which has a horizontal asymptote?

A. $(x + 2)(x - 1)(x - 3)$

B. $x^4 - 4x^2$

C. $x(12 - 2x)^2$

D. $(100 + 2x)/x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q4/5

Explain how to compare end behavior of a polynomial given by formula and a rational function given by a graph.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.C.9 • Set 42 • FORWARD 2 • Q5/5

SET 43 • HSF-IF.B.4 • INTERPRETING FUNCTIONS

Interpret periodic models from graphs and tables

MAIN	Algebra II main question	**HSF-IF.B.4**
BACKWARD 1	Interpret maximums and minimums on a graph	**HSF-IF.B.4**
BACKWARD 2	Read values from a table	**6.EE.C.9**
BACKWARD 3	Amplitude, period, and midline of a sinusoidal model	**HSF-TF.B.5**
FORWARD 1	Predict with periodicity	**HSF-IF.B.4**
FORWARD 2	Write a model from periodic features	**HSF-TF.B.5**

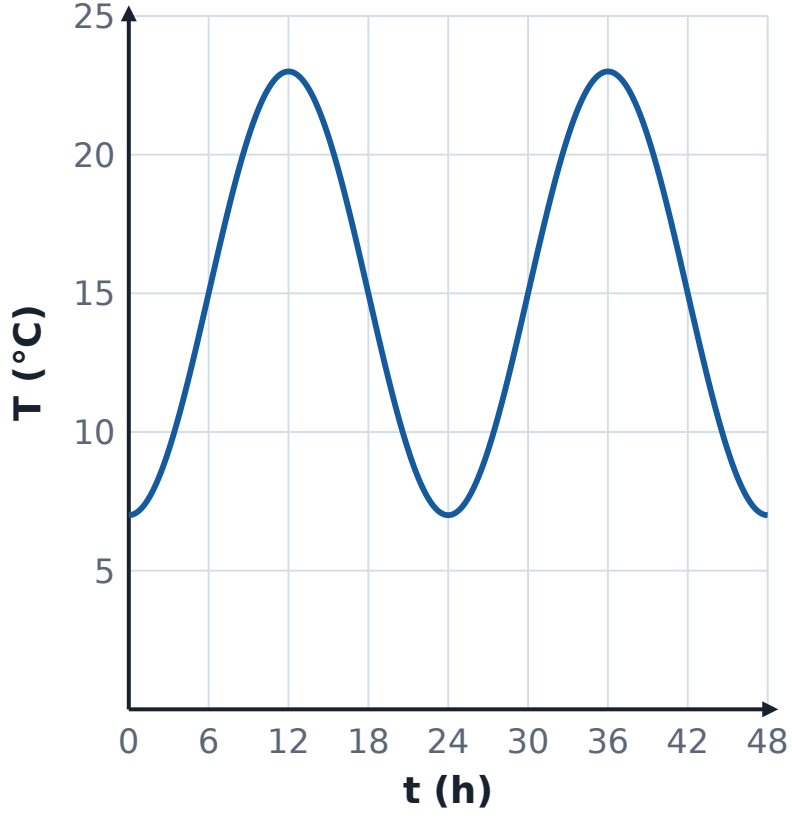
SET 43 • HSF-IF.B.4

MAIN

Algebra II main question • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows a city's temperature T (°C) t hours after midnight on two days. Give the period, the maximum temperature, and the first time it occurs.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q1/5

The table shows the water depth d (ft) at a harbor every 2 hours. Give the period, the depth at low tide, and the first time of low tide.

t (h)	0	2	4	6	8	10	12	14	16	18
d (ft)	9	7	3	1	3	7	9	7	3	1

Answer: _____

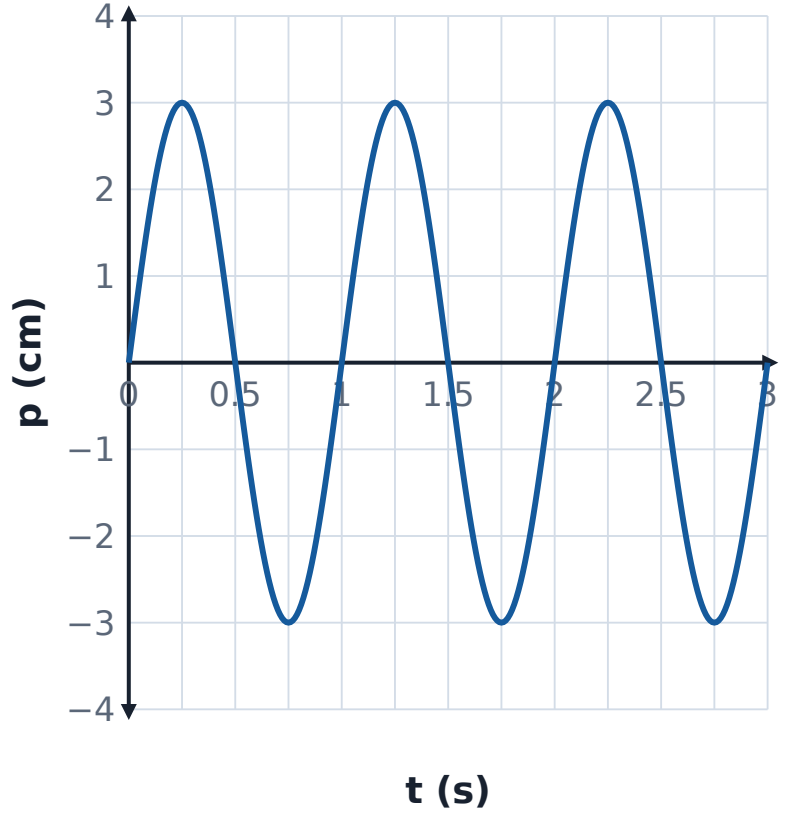
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q2/5

A piston's position p (cm) is graphed for $0 \le t \le 3$ seconds. Give the amplitude, the period, and the intervals of the first cycle where p increases.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q3/5

The table shows hours of daylight D in month m (m = 0 is January). Give the maximum, the minimum, the period, and the months in which daylight increases.

m	0	1	2	3	4	5	6	7	8	9	10	11	12
D (h)	9	9.4	10.5	12	13.5	14.6	15	14.6	13.5	12	10.5	9.4	9

Answer: _____

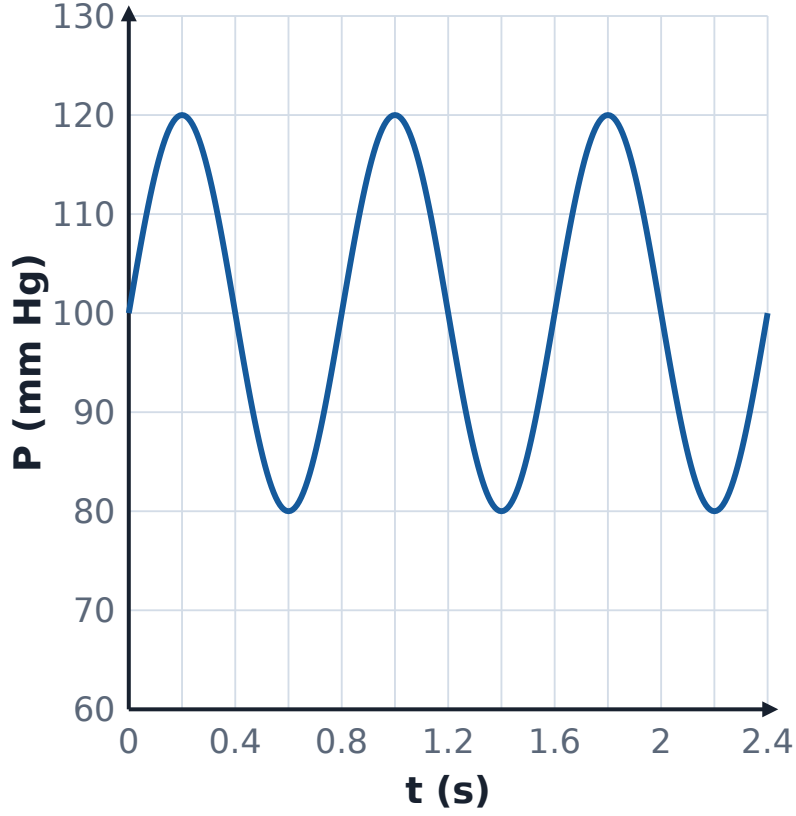
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q4/5

The graph shows blood pressure P (mm Hg) over 2.4 seconds. Give the maximum and minimum pressures and the number of heartbeats per minute.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • MAIN • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 1

Interpret maximums and minimums on a graph • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4

A graph of height peaks at 45 m. What does the peak mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q1/5

A graph is lowest at (6, 1). Interpret it if it shows depth (ft) against time (h).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q2/5

True or false?

The minimum value of a graph is its lowest y-value.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q3/5

A relative maximum on a graph is

A. the last point

B. any x-intercept

C. a point higher than nearby points

D. the y-intercept

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q4/5

A graph rises on $0 < x < 2$. What does this mean for a temperature graph?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 2

Read values from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. What is d when t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q1/5

Harbor depths d (ft) at t = 0, 2, 4, 6, 8, 10, 12 hours are 9, 7, 3, 1, 3, 7, 9. At which listed times is d = 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q2/5

True or false?

A repeated pattern in a table suggests a periodic relationship.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q3/5

Hours of daylight in months m = 0, 3, 6, 12 are 9, 12, 15, 9. Daylight of 15 h occurs at

A. m = 12

B. m = 6

C. m = 0

D. m = 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q4/5

Harbor depth is 7 ft at t = 2 h and 3 ft at t = 4 h. By how much does the depth d change from t = 2 to t = 4?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • HSF-IF.B.4

BACKWARD 3

Amplitude, period, and midline of a sinusoidal model • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Give the amplitude and midline of $T = 15 - 8\cos(\pi t/12)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q1/5

Give the period of $p = 3\sin(2\pi t)$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q2/5

True or false?

A wave between 80 and 120 has amplitude 40.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q3/5

A wave with maximum 9 and minimum 1 has midline

A. $y = 8$

B. $y = 5$

C. $y = 10$

D. $y = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q4/5

A period of 0.8 s means how many cycles per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • HSF-IF.B.4

FORWARD 1

Predict with periodicity • High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4

Use the temperature model $T(t) = 15 - 8\cos(\pi t/12)$ to find T at t = 100 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q1/5

Harbor depths repeat every 12 hours, and the depth is 1 ft at t = 6 h. Predict the depth at t = 30 hours.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q2/5

True or false?

If f has period 12, then f(50) = f(10).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q3/5

A piston's position is $p = 3\sin(2\pi t)$ cm, with period 1 s. At $t = 7.25$ s, p is

A. 0

B. 3

C. -3

D. 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F1-Q4**

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q4/5

Daylight follows a 12-month cycle with a maximum of 15 h at m = 6 (m = 0 is January). In which month after m = 12 does daylight next reach 15 h?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School — Algebra II, revisits an Algebra I standard • Functions: Interpreting Functions • HSF-IF.B.4 • Set 43 • FORWARD 1 • Q5/5

SET 43 • HSF-IF.B.4

FORWARD 2

Write a model from periodic features • High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5

Harbor depth varies between 9 ft (at t = 0) and 1 ft (at t = 6 h) and repeats every 12 hours. Write a cosine model for the depth d.

d = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q1/5

Daylight varies between 9 h (month m = 0) and 15 h (m = 6) and repeats every 12 months. Write a model for daylight D.

D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

High School — Algebra II • Functions: Trigonometric Functions • HSF-TF.B.5 • Set 43 • FORWARD 2 • Q2/5